# SKEW-SYMMETRY AND CONTRAGREDIENT TRANSFORMATIONS OF GENERAL TWISTED INTERTWINING OPERATORS

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*Allowed standard prerequisites (editorial).*

- Vertex operator algebras, twisted modules and ordinary complex analytic continuation. Complete original bibliography and source-local definitions retained.

*Source reference.* Original reference 4: algebra-geometry/sources/sigma-2026-086/native/sigma26-086.tex, source lines 2738–2738. Pointer to another section or introductory illustration; complete target construction and source-local proof support are retained.

*Source reference.* Original reference 5: algebra-geometry/sources/sigma-2026-086/native/sigma26-086.tex, source lines 3539–3539. Pointer to another section or introductory illustration; complete target construction and source-local proof support are retained.

## 1. Twisted modules and twisted intertwining operators

We first recall in this section the notion of (generalized) twisted module from [15]. We then introduce a notion of twisted intertwining operators more general than the one in [17]. We also give some basic results on such twisted intertwining operators.

For $z \in \mathbb{C}^\times$ and $p \in \mathbb{Z}$, we shall use the notation $l_p(z) = \log|z| + \mathrm{i}\arg z + 2\pi p\mathrm{i}$, where $0 \le \arg z < 2\pi$. We shall also use the notation $\log z = l_0(z) = \log|z| + \mathrm{i}\arg z$. For a vector space $U$, $p \in \mathbb{Z}$ and a formal series

$$f(x) = \sum_{k=0}^{K} \sum_{n\in\mathbb{C}} a_{n,k} x^n (\log x)^k,$$

where $a_{n,k} \in U$, the series

$$f^p(z) = \sum_{k=0}^{K} \sum_{n\in\mathbb{C}} a_{n,k} \mathrm{e}^{nl_p(z)} (l_p(z))^k$$

is called the $p$-th analytic branch of $f(x)$. We also denote $f^0(z)$ simply by $f(z)$.

Let $V$ be a vertex operator algebra and $g$ an automorphism of $V$. We recall the definition of generalized $g$-twisted $V$-module first introduced in [15]. Note that as in [15, 17] in this paper, the vertex operator map for a generalized $g$-twisted $V$-module in general contains the logarithm of the variable and the operator $L(0)$ in general does not have to act semisimply.

**Definition 1.1.** *A generalized $g$-twisted $V$-module without a $g$-action is a $\mathbb{C}$-graded vector space $W = \coprod_{n\in\mathbb{C}} W_{[n]}$ (graded by weights) equipped with a linear map*

$$Y_W^g : \ V \otimes W \to W\{x\}[\log x],$$

$$v \otimes w \mapsto Y_W^g(v,x)w = \sum_{k=0}^{K} \sum_{n\in\mathbb{C}} (Y_W^g)_{n,k}(v) w x^{-n-1} (\log x)^k$$

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*satisfying the following conditions:*

(1) *The equivariance property: For* $p \in \mathbb{Z}$, $z \in \mathbb{C}^\times$, $v \in V$ *and* $w \in W$, $\big(Y_W^g\big)^{p+1}(gv, z)w = \big(Y_W^g\big)^p(v, z)w$, *where for* $p \in \mathbb{Z}$, $\big(Y_W^g\big)^p(v, z)$ *is the* $p$*-th analytic branch of* $Y_W^g(v, x)$.

(2) *The identity property: For* $w \in W$, $Y_W^g(\mathbf{1}, x)w = w$.

(3) *The duality property: For any* $u, v \in V$, $w \in W$ *and* $w' \in W'$, *there exists a maximally-extended multivalued analytic function with preferred branch of the form*

$$f(z_1, z_2) = \sum_{i,j,k,l=0}^{N} a_{ijkl} z_1^{m_i} z_2^{n_j} (\log z_1)^k (\log z_2)^l (z_1 - z_2)^{-t}$$

*for* $N \in \mathbb{N}$, $m_1, \dots, m_N$, $n_1, \dots, n_N \in \mathbb{C}$ *and* $t \in \mathbb{Z}_+$, *such that the series*

$$\big\langle w', \big(Y_W^g\big)^p(u, z_1)\big(Y_W^g\big)^p(v, z_2)w\big\rangle = \sum_{n\in\mathbb{C}} \big\langle w', \big(Y_W^g\big)^p(u, z_1)\pi_n\big(Y_W^g\big)^p(v, z_2)w\big\rangle,$$

$$\big\langle w', \big(Y_W^g\big)^p(v, z_2)\big(Y_W^g\big)^p(u, z_1)w\big\rangle = \sum_{n\in\mathbb{C}} \big\langle w', \big(Y_W^g\big)^p(v, z_2)\pi_n\big(Y_W^g\big)^p(u, z_1)w\big\rangle,$$

$$\big\langle w', \big(Y_W^g\big)^p(Y_V(u, z_1 - z_2)v, z_2)w\big\rangle = \sum_{n\in\mathbb{C}} \big\langle w', \big(Y_W^g\big)^p(\pi_n Y_V(u, z_1 - z_2)v, z_2)w\big\rangle$$

*are absolutely convergent on the regions* $|z_1| > |z_2| > 0$, $|z_2| > |z_1| > 0$, $|z_2| > |z_1 - z_2| > 0$, *respectively, and their sums are equal to the branch*

$$f^{p,p}(z_1, z_2) = \sum_{i,j,k,l=0}^{N} a_{ijkl} \mathrm{e}^{m_i l_p(z_1)} \mathrm{e}^{n_j l_p(z_2)} l_p(z_1)^k l_p(z_2)^l (z_1 - z_2)^{-t}$$

*of* $f(z_1, z_2)$ *on the region* $|z_1| > |z_2| > 0$, *the region* $|z_2| > |z_1| > 0$, *the region given by* $|z_2| > |z_1 - z_2| > 0$ *and* $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$, *respectively.*

(4) *The* $L(0)$*-grading condition: Let* $L_W^g(0) = \operatorname{Res}_x x Y_W^g(\omega, x)$. *Then for* $n \in \mathbb{C}$, $w \in W_{[n]}$, *there exists* $K \in \mathbb{Z}_+$ *such that* $\big(L_W^g(0) - n\big)^K w = 0$.

(5) *The* $L(-1)$*-derivative property: For* $v \in V$,

$$\frac{\mathrm{d}}{\mathrm{d}x} Y_W^g(v, x) = Y_W^g(L_V(-1)v, x).$$

*A lower-bounded generalized* $g$*-twisted* $V$*-module is a generalized* $g$*-twisted* $V$*-module* $W$ *such that for each* $n \in \mathbb{C}$, $W_{[n+l]} = 0$ *for sufficiently negative real numbers* $l$. *A generalized* $g$*-twisted* $V$*-module* $W$ *is said to be grading-restricted if it is lower bounded and for each* $n \in \mathbb{C}$, $\dim W_{[n]} < \infty$. *A generalized* $g$*-twisted* $V$*-module with a* $g$*-action is a* $\mathbb{C} \times \mathbb{C}/\mathbb{Z}$*-graded vector space* $W = \coprod_{n\in\mathbb{C},\alpha\in\mathbb{C}/\mathbb{Z}} W_{[n]}^{[\alpha]}$ *(graded by weights and* $g$*-weights) equipped with a linear map* $Y_W^g : V \otimes W \to W\{x\}[\log x]$ *as above and an action of* $g$ *such that* $W$ *equipped with* $Y_W^g$ *is a generalized* $g$*-twisted* $V$*-module without a* $g$*-action satisfying in addition the following* $g$*-grading condition: There exists* $\Lambda \in \mathbb{Z}_+$ *such that for* $\alpha \in \mathbb{C}/\mathbb{Z}$, $w \in W^{[\alpha]} = \coprod_{n\in\mathbb{N}} W_{[n]}^{[\alpha]}$, $\big(g - \mathrm{e}^{2\pi\alpha\mathrm{i}}\big)^\Lambda w = 0$. *Moreover,* $gY_W^g(u, x)w = Y_W^g(gu, x)gw$ *for* $u \in V$ *and* $w \in W$.

Module maps between $g$-twisted $V$-modules are defined in the obvious way. For two different automorphisms $g_1$ and $g_2$ of $V$, there might exist linear maps from a $g_1$-twisted $V$-module to a $g_2$-twisted $V$-module commuting with the twisted vertex operator maps when $V$ is not simple. But we will not consider such maps in this paper. So in the categories of twisted $V$-modules considered in this paper, we always choose the space of module maps between a $g_1$-twisted $V$-module and a $g_2$-twisted $V$-module to be 0 if $g_1 \neq g_2$.

In this paper, generalized $g$-twisted $V$-modules with or without $g$-actions are all called *generalized* $g$*-twisted* $V$*-modules*. Only when the $g$-actions are specifically

discussed, we shall add the words "with a $g$-action" or "without a $g$-action". Later, we shall also need generalized $g$-twisted $V$-modules with $g$-actions equipped with an $h$-action for another automorphism $h$ of $V$. Also, for simplicity, we shall sometimes omit the subscript $W$ to denote the twisted vertex operator map $Y_W^g$ by $Y^g$.

**Remark 1.2.** Note that a lower-bounded generalized $g$-twisted $V$-module defined above also satisfies the lower-truncation property: For $v \in V$ and $w \in W$, $n \in \mathbb{C}$ and $k = 0, \dots, K$, $\big(Y_W^g\big)_{n+l,k}(w_1)w_2 = 0$ for $l \in \mathbb{N}$ sufficiently large. This in fact follows from the $L(0)$-commutator formula, which is a consequence of the duality property and the $L(0)$-grading condition.

**Remark 1.3.** In this paper, we need to study generalized $g$-twisted $V$-modules with and without $g$-actions in a different way because the skew-symmetry and contragredient isomorphisms in Section 2 are different in these two types of twisted modules. In fact, there is also a notion of generalized $g$-twisted module with a $G$ action for a group $G$ of automorphisms of $V$. See [22] for a definition and the construction $P(z)$-tensor product of such twisted modules.

In this paper, we shall consider only lower-bounded generalized $g$-twisted $V$-modules and grading-restricted generalized $V$-modules. For simplicity, we shall often call lower-bounded generalized $g$-twisted $V$-modules simply $g$-twisted $V$-modules and grading-restricted generalized $V$-modules simply grading-restricted $g$-twisted $V$-modules.

Let $\big(W, Y_W^g\big)$ be a $g$-twisted $V$-module without a $g$-action. Let $h$ be an automorphism of $V$. We recall the $hgh^{-1}$-twisted $V$-module $\big(W, \phi_h\big(Y_W^g\big)\big)$ without a $hgh^{-1}$-action (see, for example, [17]). Let

$$\phi_h\big(Y_W^g\big)\colon\ V \times W \to W\{x\}[\log x], \qquad v \otimes w \mapsto \phi_h\big(Y_W^g\big)(v,x)w$$

be the linear map defined by

$$\phi_h\big(Y_W^g\big)(v,x)w = Y_W^g\big(h^{-1}v, x\big)w.$$

Then the pair $\big(W, \phi_h\big(Y_W^g\big)\big)$ is an $hgh^{-1}$-twisted $V$-module without a $hgh^{-1}$-action.

In the case that $W$ is a $g$-twisted $V$-module with a $g$-action and also has an action of $h$ in the sense that $W$ is a module for the group generated by $g$ and $h$, we define

$$\varphi_h\big(Y_W^g\big)\colon\ V \times W \to W\{x\}[\log x], \qquad v \otimes w \mapsto \varphi_h\big(Y_W^g\big)(v,x)w$$

by

$$\varphi_h\big(Y_W^g\big)(v,x)w = hY_W^g\big(h^{-1}v, x\big)h^{-1}w.$$

Then $\big(W, \varphi_h\big(Y_W^g\big)\big)$ is also an $hgh^{-1}$-twisted $V$-module without a $hgh^{-1}$-action. Since the conformal element $\omega$ is fixed under the action of $g^{-1}$, the coefficients

$$L_{\varphi_h(W)}(n) = \operatorname{Res}_x x^{n+1}\varphi_h\big(Y_W^g\big)(\omega,x) = h\operatorname{Res}_x x^{n+1}Y_W^g(\omega,x)h^{-1} = hL_W^g(n)h^{-1}$$

of the vertex operator $\varphi_h\big(Y_W^g\big)(\omega,x)$ for $n \in \mathbb{N}$ also give Virasoro operators on $W$ with the same central charge. Since $h$ is an isomorphism of the vector space $W$,

$$hW = \coprod_{n\in\mathbb{C},\alpha\in\mathbb{C}/\mathbb{Z}} hW_{[n]}^{[\alpha]}.$$

We denote the graded space $hW$ by $\varphi_h(W)$ and $hW_{[n]}^{[\alpha]}$ by $\varphi_h(W)_{[n]}^{[\alpha]}$ $n \in \mathbb{C}, \alpha \in \mathbb{C}/\mathbb{Z}$. Then we have

$$\varphi_h(W) = \coprod_{n\in\mathbb{C},\alpha\in\mathbb{C}/\mathbb{Z}} \varphi_h(W)_{[n]}^{[\alpha]}.$$

Note that as a vector space $\varphi_h(W) = hW$ is the same as $W$, but as graded spaces, they are different. For $w \in \varphi_h(W)_{[n]} = \coprod_{\alpha\in\mathbb{C}/\mathbb{Z}} \varphi_h(W)^{[\alpha]}_{[n]}$, $h^{-1}w \in W_{[n]} = \coprod_{\alpha\in\mathbb{C}/\mathbb{Z}} W^{[\alpha]}_{[n]}$. So there exists $K \in \mathbb{Z}_+$ such that $\big(L^g_W - n\big)^K h^{-1}w = 0$. Then

$$(L_{\varphi_h(W)}(0) - n)^K w = h\big(L^g_W - n\big)^K h^{-1}w = 0.$$

For $w \in \varphi_h(W)^{[\alpha]} = \coprod_{n\in\mathbb{Z}} \varphi_h(W)^{[\alpha]}_{[n]}$, $h^{-1}w \in W^{[\alpha]} = \coprod_{n\in\mathbb{Z}} W^{[\alpha]}_{[n]}$. So there exists $\Lambda \in \mathbb{Z}_+$ such that $\big(g - \mathrm{e}^{2\pi\mathrm{i}\alpha}\big)^\Lambda h^{-1}w = 0$. Then $\big(hgh^{-1} - \mathrm{e}^{2\pi\mathrm{i}\alpha}\big)^\Lambda w = h\big(g - \mathrm{e}^{2\pi\mathrm{i}\alpha}\big)^\Lambda h^{-1}w = 0$. Also we have

$$\begin{aligned} hgh^{-1}\varphi_h\big(Y^g_W\big)(v,x)w &= hgY^g_W\big(h^{-1}v,x\big)h^{-1}w = hY^g_W\big(h^{-1}hgh^{-1}v,x\big)h^{-1}hgh^{-1}w \\ &= \varphi_h\big(Y^g_W\big)\big(hgh^{-1}v,x\big)hgh^{-1}w. \end{aligned}$$

So $(\varphi_h(W), \varphi_h\big(Y^g_W\big))$ is a $hgh^{-1}$-twisted $V$-module with an $hgh^{-1}$-action. We shall denote this $hgh^{-1}$-twisted $V$-module with an $hgh^{-1}$-action by $\varphi_h(W)$.

Note that when $h = g$, $\varphi_h(W) = W$ and $\phi_g(W)$ with the twisted vertex operator given by

$$\phi_g\big(Y^g_W\big)(v,x)w = Y^g_W\big(g^{-1}v,x\big)w$$

is equivalent to the original $g$-twisted module $W$ with the equivalence $g^{-1}\colon W \to W$. By the equivariance property, we have $\big(Y^g_W\big)^p\big(g^{-1}v,x\big) = \big(Y^g_W\big)^{p+1}(v,x)$ and, if

$$Y^g_W(v,x)w = \sum_{k=0}^{K}\sum_{n\in\mathbb{C}} \big(Y^g_W\big)_{n,k}(v)x^{-n-1}(\log x)^k$$

for $v \in V$ and $w \in W$, we have

$$Y^g_W\big(g^{-1}v,x\big)w = \sum_{k=0}^{K}\sum_{n\in\mathbb{C}} \big(Y^g_W\big)_{n,k}(v)\mathrm{e}^{-2\pi\mathrm{i}n}x^{-n-1}(\log x + 2\pi\mathrm{i})^k.$$

**Remark 1.4.** In the definitions of $\big(\varphi_h(W), \varphi_h\big(Y^g_W\big)\big)$, we assume that there is an action of $h$ on $W$. One immediate question is whether there exist such $g$-twisted $V$-modules with $h$-actions. In fact, the same construction in Section 5 of [18] can be used to construct such a $g$-twisted $V$-module with an $h$-action or even a $g$-twisted $V$-module with a $G$-action for a group $G$ of automorphisms of $V$ satisfying the additional condition $hY^g_W(v,x)w = Y^g_W(hv,x)hw$ for $h \in G$. In this case, though $\varphi_h(W) = W$ as vector spaces and $\varphi_h\big(Y^g_W\big) = Y^g_W$ as linear maps from $V \otimes W$ to $W\{x\}[\log x]$ when we view $\varphi_h(W)$ as a vector space equal to $W$, $\varphi_h(W)$ is graded by generalized eigenspaces of $hgh^{-1}$ but $W$is graded by the generalized eigenspaces of $g$. In particular, $\big(\varphi_h(W), \varphi_h\big(Y^g_W\big)\big)$ and $\big(W, Y^g_W\big)$ are indeed different twisted $V$-modules.

We also recall contragredient twisted $V$-modules (see, for example, [17]). Let $\big(W, Y^g_W\big)$ be a $g$-twisted $V$-module relative to $G$. Let $W'$ be the graded dual of $W$. Define a linear map

$$\big(Y^g_W\big)'\colon\ V \otimes W' \to W'\{x\}[\log x], \qquad v \otimes w' \mapsto \big(Y^g_W\big)'(v,x)w'$$

by

$$\big\langle \big(Y^g_W\big)'(v,x)w', w\big\rangle = \big\langle w', Y^g_W\big(\mathrm{e}^{xL(1)}\big(-x^{-2}\big)^{L(0)}v, x^{-1}\big)w\big\rangle$$

for $v \in V$, $w \in W$ and $w' \in W'$. Then the pair $\big(W', \big(Y^g_W\big)'\big)$ is a $g^{-1}$-twisted $V$-module.

Let $M^2 = \big\{(z_1,z_2) \in \mathbb{C}^2 \mid z_1 \neq 0,\ z_2 \neq 0,\ z_1 \neq z_2\big\}$. Let $f(z_1,z_2)$ be a maximally extended multivalued analytic function on $M^2$ with a preferred single-valued branch

$f^e(z_1, z_2)$ on the simply-connected region $M_0^2$ given by cutting $M^2$ along the positive real lines in the $z_1$-, $z_2$- and $(z_1 - z_2)$-planes, that is, the sets

$$\{(z_1, z_2) \in M^2 \mid z_1 \in \mathbb{R}_+\}, \qquad \{(z_1, z_2) \in M^2 \mid z_2 \in \mathbb{R}_+\},$$
$$\{(z_1, z_2) \in M^2 \mid z_1 - z_2 \in \mathbb{R}_+\},$$

with these sets attached to the upper half $z_1$-, $z_2$- and $(z_1 - z_2)$-planes, that is, the points in these sets are viewed as limits in $M_0^2$ of sequences of points approaching them from the upper $z_1$-, $z_2$- and $(z_1 - z_2)$-planes. Note that given any point $(z_1^0, z_2^0) \in M^2$ and any loop $\gamma$ in $M^2$ based at $(z_1^0, z_2^0)$, we obtain from the preferred branch of $f(z_1, z_2)$ another single-valued branch by going around $\gamma$. The resulting single-valued branch depends only on the homotopy class of the loop and is independent of the choices of the base point. Thus we obtain a right action of the fundamental group of $M^2$ on the set of single-valued branches of $f(z_1, z_2)$. Note that $M^2$ is homotopically equivalent to the configuration space

$$F_3(\mathbb{C}) = \{(z_1, z_2, z_3) \in \mathbb{C}^3 \mid z_i \neq z_j,\ i \neq j\}.$$

So the fundamental group of $M^2$ is in fact the pure braid group $\mathrm{PB}_3$ and has three generators $b_{12}$, $b_{13}$ and $b_{23}$, which are given as follows: Choose the base point to be $(-3, -2)$. Then the generator $b_{12}$ is the homotopy class of the loop given by letting $z_1$ go counterclockwise around the circle of radius 1 centered at $-2$ (see Figure 1). The generator $b_{23}$ is the homotopy class of the loop given by letting $z_2$ go counterclockwise around the circle of radius 2 centered at 0 (see Figure 2). The generator $b_{13}$ is the homotopy class of the loop given by letting $z_1$ go around first the lower half circle of radius 3 centered at 0 counterclockwise, then the upper half circle of radius 2 centered at 1 counterclockwise and finally the lower half circle of radius 1 centered at $-2$ clockwise (see Figure 3). We know that the pure braid group $\mathrm{PB}_3$ is isomorphic to the group generated by $b_{12}$, $b_{13}$, $b_{23}$ with the relations $b_{13}b_{12}b_{23} = b_{12}b_{23}b_{13} = b_{23}b_{13}b_{12}$. See [4] for more detailed discussions on $M^2$, its fundamental group, the pure braid groups and multivalued functions on $M^2$.

For $a \in \mathrm{PB}_3$, we denote the single-valued branch of $f(z_1, z_2)$ obtained by applying $a$ to $f^e(z_1, z_2)$ by $f^a(z_1, z_2)$.

Let $\delta = (\delta_1, \dots, \delta_n)$, where $\delta_i \in \{0, \infty\}$, $i = 1, \dots, n$. Suppose $f(z_1, \dots, z_n)$ is a multi-valued analytic function defined on an open region $\Omega$ of $\mathbb{C}^n$. We say $(z_1, \dots, z_n) = \delta$ is a *component-isolated singularity of* $f(z_1, \dots, z_n)$ if there exists $r = (r_1, \dots, r_n) \in \mathbb{R}_+^n$ such that $\Delta^\times(\delta, r) \subset \Omega$, where $\Delta^\times(\delta, r) = \{(z_1, \dots, z_n) \in \mathbb{C}^n \mid |z_i| > r_i \text{ if } \delta_i = \infty, 0 < |z_i| < r_i \text{ if } \delta_i = 0\}$. Let $A \in \mathrm{GL}(n, \mathbb{C})$ and $\beta \in \mathbb{C}^n$ (written as a row vector). Then $\zeta_1, \dots, \zeta_n$ given by

$$(\zeta_1, \dots, \zeta_n) = (z_1, \dots, z_n)A - \beta$$

are also independent variables. Define

$$g(\zeta_1, \dots, \zeta_n) = f\left((\zeta_1, \dots, \zeta_n)A^{-1} + \beta A^{-1}\right). \tag{1.1}$$

For $\delta \in (\mathbb{C} \cup \{\infty\})^n$, we say that $(\zeta_1, \dots, \zeta_n) = \delta$ is a *component-isolated singularity of the function* $f(z_1, \dots, z_n)$ if $(\zeta_1, \dots, \zeta_n) = \delta$ is a component-isolated singularity of $g(\zeta_1, \dots, \zeta_n)$.

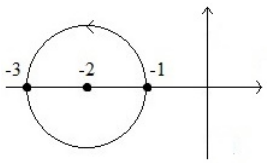



FIGURE 1. The loop for $b_{12}$.

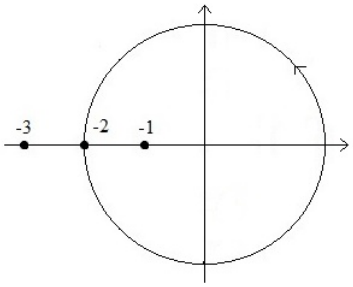



FIGURE 2. The loop for $b_{23}$.

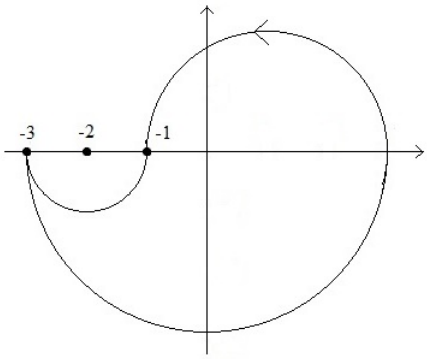



FIGURE 3. The loop for $b_{13}$.

**Remark 1.5.** Notice that $(\zeta_1, \dots, \zeta_n)(= (z_1, \dots, z_n)A - \beta) = \delta$ being a component-isolated singularity of a function is not equivalent to $(z_1, \dots, z_n) = \delta A^{-1} + \beta A^{-1}$ being a component-isolated singularity of the same function. This is because we have different sets of independent variables. For example, consider $f(z_1, z_2) = \frac{1}{z_1 - z_2}$. Since $(\zeta_1, \zeta_2) = (0, 0)$ is a component-isolated singularity of the function $g(\zeta_1, \zeta_2) = 1/\zeta_1$, we also say that $(z_1 - z_2, z_2) = (0, 0)$ is a component-isolated singularity of $f(z_1, z_2)$. In this case, $z_1 - z_2$ and $z_2$ are independent variables. However, $(z_1, z_2) = (0, 0)$ is clearly not a component-isolated singularity of $f(z_1, z_2)$. In this case, the independent variables are $z_1$ and $z_2$.

**Definition 1.6.** Let $f(z_1, \dots, z_n)$ be a multi-valued analytic function defined on an open region of $\mathbb{C}^n$. Let $\delta = (\delta_1, \dots, \delta_n) \in \{0, \infty\}^n$. Suppose $(z_1, \dots, z_n) = \delta$ is a component-isolated singularity of $f(z_1, \dots, z_n)$. Let $\{f^b(z_1, \dots, z_n)\}_{b \in B}$ be the set of all single valued branches of $f(z_1, \dots, z_n)$ near $\delta$ with cuts at $z_i \in \mathbb{R}_+$. Then for each $b \in B$, there exists $r_b \in \mathbb{R}_+^n$ such that $f^b(z_1, \dots, z_n)$ is analytic on $\Delta^\times(\delta, r_b)$. We say that $(z_1, \dots, z_n) = \delta$ is a *regular singularity* of $f(z_1, \dots, z_n)$ if there exists $K \in \mathbb{N}$, $D_i = \cup_{j=1}^{N_i} r_j^{(i)} + \mathbb{N}$ $\big($or $D_i = \cup_{j=1}^{N_i} r_j^{(i)} - \mathbb{N}\big)$ where $r_1^{(i)}, \dots, r_{N_i}^{(i)} \in \mathbb{C}$ for $\delta_i = 0$ (or $\delta_i = \infty$), and $\alpha^{(b)}_{a_1, j_1; \dots; a_n, j_n} \in \mathbb{C}$, such that on the region $\Delta^\times(\delta, r_b)$, the right-hand-side of the following equation is absolutely convergent, and

(1.2)

$$f^b(z_1, \dots, z_n) = \sum_{i=1}^{n} \sum_{a_i \in D_i} \sum_{j_1, \dots, j_n = 0}^{K} \alpha^{(b)}_{a_1, j_2; \dots; a_n, j_n} z_1^{a_1} (\log z_1)^{j_1} \cdots z_n^{a_n} (\log z_n)^{j_n}.$$

If $K = 0$ and $D_i = -n^{(i)} + \mathbb{N}$ in the case $\delta_i = 0$ and $D_i = n^{(i)} - \mathbb{N}$ in the case $\delta_i = \infty$ for $i = 1, \dots, n$ in ((1.2)), where $n^{(i)} \in \mathbb{Z}_+$ for $i \in I \subset \{1, \dots, n\}$ and

$n^{(i)} = 0$ for $i \neq I$, we say that $z_i = \delta_i$ for $i \in I$ are *poles of* $f(z_1, \dots, z_n)$. If $n^{(i)} \in \mathbb{Z}_+$ for $i \in I \subset \{1, \dots, n\}$, $\delta_i = 0$ are the smallest and such that (1.2) holds, we call $n_i$ the orders of the poles $z_i = \delta_i$, respectively, for $i \in \mathbb{I}$. Let $A \in \mathrm{GL}(n, \mathbb{C})$ and $\beta \in \mathbb{C}^n$ be the same as above. We say that $(\zeta_1, \dots, \zeta_n) = \delta$ is a *regular singularity* of $f(z_1, \dots, z_n)$ if $(\zeta_1, \dots, \zeta_n) = \delta$ is a regular singularity of $g(\zeta_1, \dots, \zeta_n)$, where $g(\zeta_1, \dots, \zeta_n)$ is given by (1.1). We say that $\zeta_i = \delta_i$ for $i \in I$ are poles of $f(z_1, \dots, z_n)$ with orders $n_i$, respectively, if $\zeta_i = \delta_i$ for $i \in I$ are poles of $g(\zeta_1, \dots, \zeta_n)$ with orders $n_i$, respectively.

**Remark 1.7.** Let $f(z_1, \dots, z_n)$ and $g(z_1, \dots, z_n)$ be multivalued analytic functions with preferred branches defined on an open region. Then $\lambda f(z_1, \dots, z_n) + \mu g(z_1, \dots, z_n)$ for $\lambda, \mu \in \mathbb{C}$ and $f(z_1, \dots, z_n)g(z_1, \dots, z_n)$ are well defined using the preferred branches and are also multivalued analytic functions on the same region with preferred branches. If $(\zeta_1, \dots, \zeta_n) = ((z_1, \dots, z_n)A - \beta =)\delta$ is a regular singular point of both $f(z_1, \dots, z_n)$ and $g(z_1, \dots, z_n)$, then it is also a regular singular point for $\lambda f(z_1, \dots, z_n) + \mu g(z_1, \dots, z_n)$ and $f(z_1, \dots, z_n)g(z_1, \dots, z_n)$. Therefore, the set of multivalued analytic functions on the same region with a preferred branch such that $(\zeta_1, \dots, \zeta_n) = \delta$ is a regular singular point form a commutative associative algebra over $\mathbb{C}$.

We also need the region $M^n = \{(z_1, \dots, z_n) \in \mathbb{C}^n \mid z_i \neq 0, z_i \neq z_j, i \neq j\}$ for $n \in \mathbb{Z}_+$.

**Definition 1.8.** *Let $g_1$, $g_2$, $g_3$ be automorphisms of $V$ and let $W_1$, $W_2$ and $W_3$ be $g_1$-, $g_2$- and $g_3$-twisted $V$-modules (with or without actions of the corresponding automorphisms of $V$), respectively. A twisted intertwining operator of type $\binom{W_3}{W_1 W_2}$ is a linear map*

$$\mathcal{Y}: \ W_1 \otimes W_2 \to W_3\{x\}[\log x],$$

$$w_1 \otimes w_2 \mapsto \mathcal{Y}(w_1, x)w_2 = \sum_{k=0}^{K} \sum_{n \in \mathbb{C}} \mathcal{Y}_{n,k}(w_1)w_2 x^{-n-1}(\log x)^k$$

*satisfying the following conditions:*

1. *The lower truncation property: For $w_1 \in W_1$ and $w_2 \in W_2$, $n \in \mathbb{C}$ and $k = 0, \dots, K$, $\mathcal{Y}_{n+l,k}(w_1)w_2 = 0$ for $l \in \mathbb{N}$ sufficiently large.*
2. *The duality property: For $u \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w_3' \in W_3'$, there exists a maximally extended multivalued analytic function $f(z_1, z_2; u, w_1, w_2, w_3')$ on $M^2$ with a preferred single-valued branch $f^e(z_1, z_2; u, w_1, w_2, w_3')$ on $M_0^2$ such that the series*

$$\langle w_3', Y_{W_3}^{g_3}(u, z_1)\mathcal{Y}(w_1, z_2)w_2\rangle = \sum_{n \in \mathbb{C}} \langle w_3', Y_{W_3}^{g_3}(u, z_1)\pi_n \mathcal{Y}(w_1, z_2)w_2\rangle, \tag{1.3}$$

$$\langle w_3', \mathcal{Y}(w_1, z_2)Y_{W_2}^{g_2}(u, z_1)w_2\rangle = \sum_{n \in \mathbb{C}} \langle w_3', \mathcal{Y}(w_1, z_2)\pi_n Y_{W_2}^{g_2}(u, z_1)w_2\rangle, \tag{1.4}$$

$$\langle w_3', \mathcal{Y}\big(Y_{W_1}^{g_1}(u, z_1 - z_2)w_1, z_2\big)w_2\rangle = \sum_{n \in \mathbb{C}} \langle w_3', \mathcal{Y}(\pi_n Y_{W_1}^{g_1}(u, z_1 - z_2)w_1, z_2)w_2\rangle \tag{1.5}$$

*are absolutely convergent on the regions $|z_1| > |z_2| > 0$, $|z_2| > |z_1| > 0$, $|z_2| > |z_1 - z_2| > 0$, respectively. Moreover, their sums are equal to $f^e(z_1, z_2; u, w_1, w_2, w_3')$ on the region given by $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$, the region given by $|z_2| > |z_1| > 0$ and $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$, the region given by $|z_2| > |z_1 - z_2| > 0$ and $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$, respectively.*

(3) *The convergence and analytic extension for products with more than one twisted vertex operators: For* $k \in \mathbb{N} + 3$, $u_1, \dots, u_{k-1} \in V$, $w_1 \in W_1$, $w_2 \in W_2$ *and* $w_3' \in W_3'$, *the series*

$$\begin{aligned}&\left\langle w_3', Y_{W_3}^{g_3}(u_1, z_1) \cdots Y_{W_3}^{g_3}(u_{k-1}, z_{k-1}) \mathcal{Y}(w_1, z_k) w_2 \right\rangle \\ = &\sum_{n_1, \dots, n_{k-1} \in \mathbb{C}} \left\langle w_3', Y_{W_3}^{g_3}(u_1, z_1) \pi_{n_1} \cdots \pi_{n_{k-2}} Y_{W_3}^{g_3}(u_{k-1}, z_{k-1}) \pi_{k-1} \mathcal{Y}(w_1, z_k) w_2 \right\rangle\end{aligned}$$

*is absolutely convergent on the region* $|z_1| > \cdots > |z_k| > 0$ *and can be maximally extended to a multivalued analytic function* $f(z_1, \dots, z_k)$ *on the region* $M^k$. *Moreover, for any fixed* $z_k \in \mathbb{C}^\times$ *and* $M_{ij} \in \mathbb{Z}_+$ *such that* $x^{M_{ij}} Y_V(u_i, x) u_j \in V[[x]]$ *for* $i, j = 1, \dots, k-1$, *any component-isolated singularity of the form* $(z_1, \dots, z_{k-1}) - \beta = \delta$ *for* $\beta \in \mathbb{C}^{k-1}$ *and* $\delta \in \{0, \infty\}^{k-1}$ *of the multivalued analytic function of* $z_1, \dots, z_{k-1}$ *obtained as a value at* $\zeta = z_k$ *of* $\prod_{1 \le i < j \le k-1} (z_i - z_j)^{M_{ij}} f(z_1, \dots, \zeta)$ *is regular.*

(4) *The* $L(-1)$*-derivative property* $\frac{\mathrm{d}}{\mathrm{d}x} \mathcal{Y}(w_1, x) = \mathcal{Y}(L(-1) w_1, x)$.

**Remark 1.9.** Note that the series in the definition of twisted intertwining operator $\mathcal{Y}$ above are all summed over unique expansion sets (see [29, Definition 7.5]). In fact, by the duality property and the fact that $\omega$ is fixed under $g_1$, $g_2$ and $g_3$, the $L(0)$-commutator formula for $\mathcal{Y}$ holds. Then for homogeneous $v \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w_3' \in W_3'$, the series above are in fact summed over sets of the form $(\{r_1, \dots, r_m\} + \mathbb{N}) \times \{0, \dots, K\}$, which is a unique expansion set (see [20]).

**Remark 1.10.** For simplicity, in the duality property in Definition 1.8, we use only the preferred branches of the twisted intertwining operator and the twisted vertex operators. One can derive what the products and iterates of other branches of a twisted intertwining operator converge to using the actions of the elements of $\mathrm{PB}_3$ on the single-valued branch $f^e(z_1, z_2; u, w_1, w_2, w_3')$ of the multivalued function $f(z_1, z_2; u, w_1, w_2, w_3')$ in the definition. Let $\mathcal{Y}$ be a twisted intertwining operator of type $\binom{W_3}{W_1 W_2}$. For any $p_1, p_2, p_{12} \in \mathbb{Z}$, the series

$$\left\langle w_3', \left(Y_{W_3}^{g_3}\right)^{p_1}(u, z_1) \mathcal{Y}^{p_2}(w_1, z_2) w_2 \right\rangle = \sum_{n \in \mathbb{C}} \left\langle w_3', \left(Y_{W_3}^{g_3}\right)^{p_1}(u, z_1) \pi_n \mathcal{Y}^{p_2}(w_1, z_2) w_2 \right\rangle,$$

$$\left\langle w_3', \mathcal{Y}^{p_2}(w_1, z_2) \left(Y_{W_2}^{g_2}\right)^{p_1}(u, z_1) w_2 \right\rangle = \sum_{n \in \mathbb{C}} \left\langle w_3', \mathcal{Y}^{p_2}(w_1, z_2) \pi_n \left(Y_{W_2}^{g_2}\right)^{p_1}(u, z_1) w_2 \right\rangle,$$

$$\left\langle w_3', \mathcal{Y}^{p_2}(\left(Y_{W_1}^{g_1}\right)^{p_{12}}(u, z_1 - z_2) w_1, z_2) w_2 \right\rangle = \sum_{n \in \mathbb{C}} \left\langle w_3', \mathcal{Y}^{p_2}(\pi_n \left(Y_{W_1}^{g_1}\right)^{p_{12}}(u, z_1 - z_2) w_1, z_2) w_2 \right\rangle$$

are absolutely convergent on the regions $|z_1| > |z_2| > 0$, $|z_2| > |z_1| > 0$, $|z_2| > |z_1 - z_2| > 0$, respectively. Moreover, their sums are equal to the branches

$$f^{(b_{13} b_{12})^{p_1} b_{23}^{p_2}}(z_1, z_2; u, w_1, w_2, w_3'), \qquad f^{(b_{12} b_{23})^{p_2} b_{13}^{p_1}}(z_1, z_2; u, w_1, w_2, w_3'),$$

$$f^{(b_{23} b_{13})^{p_2} b_{12}^{p_{12}}}(z_1, z_2; u, w_1, w_2, w_3'),$$

respectively, of $f(z_1, z_2; u, w_1, w_2, w_3')$ on the region given by $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$, the region given by $|z_2| > |z_1| > 0$ and $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$, the region given by $|z_2| > |z_1 - z_2| > 0$ and $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$, respectively. See [4] for more details.

**Remark 1.11.** From Definition 1.8, we see that a twisted intertwining operator of type $\binom{W_3}{W_1 W_2}$ is in fact an intertwining operator of type $\binom{W_3}{W_1 W_2}$ when $W_1$, $W_2$, and $W_3$ are viewed as (untwisted) modules for the fixed-point subalgebra of $V$ under $g_1$ and $g_2$. In particular, all the usual properties for intertwining operators holds for the vertex operators associated to elements of the fixed-point subalgebra

and the twisted intertwining operator. For example, the Jacobi identity and the commutator formula for these operators are satisfied.

**Proposition 1.12.** *Let $g_1$, $g_2$, $g_3$ be automorphisms of $V$, $W_1$, $W_2$, $W_3$ $g_1$-, $g_2$-, $g_3$-twisted generalized $V$-modules and $\mathcal{Y}$ a twisted intertwining operator of type $\binom{W_3}{W_1 W_2}$. Assume that the map $u \mapsto Y_{W_3}^{g_3}(u, x)$ is injective and $\mathcal{Y}$ is surjective in the sense that the coefficients of the series $\mathcal{Y}(w_1, x)w_2$ for $w_1 \in W_1$, $w_2 \in W_2$ span $W_3$. Then $g_3 = g_1 g_2$.*

*Proof.* By the definition of twisted intertwining operator, for $u \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w_3' \in W_3'$, there exists a multivalued analytic function $f(z_1, z_2; u, w_1, w_2, w_3')$ on $M^2$ with a preferred single-valued branch $f^e(z_1, z_2; u, w_1, w_2, w_3')$ on $M_0^2$ such that (1.3), (1.4) and (1.5) are absolutely convergent to $f^e(z_1, z_2; u, w_1, w_2, w_3')$ on the corresponding regions in given in Definition 1.8. In particular, on the region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ the sum of the series $\langle w_3', Y_{W_3}^{g_3}(g_3 u, z_1)\mathcal{Y}(w_1, z_2)w_2\rangle$ is equal to $f^e(z_1, z_2; g_3 u, w_1, w_2, w_3')$. By the equivariance property for $W_3$,

$$\langle w_3', Y_{W_3}^{g_3}(g_3 u, z_1)\mathcal{Y}(w_1, z_2)w_2\rangle = \langle w_3', \left(Y_{W_3}^{g_3}\right)^{-1}(u, z_1)\mathcal{Y}(w_1, z_2)w_2\rangle, \tag{1.6}$$

where as above $\left(Y_{W_3}^{g_3}\right)^{-1}$ is the $(-1)$-th branch of the twisted vertex operator map $Y_{W_3}^{g_3}$. But the right-hand side of (1.6) can be obtained from $\langle w_3', Y_{W_3}^{g_3}(u, z_1)\mathcal{Y}(w_1, z_2)w_2\rangle$ on the region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ by letting $z_1$ go around clockwise a circle of radius larger than $|z_2|$. The homotopy class of such a circle is equal to $(b_{13}b_{12})^{-1}$. So (1.6) gives

$$f^e(z_1, z_2; g_3 u, w_1, w_2, w_3') = f^{(b_{13}b_{12})^{-1}}(z_1, z_2; u, w_1, w_2, w_3')$$

or equivalently

$$f^{b_{13}b_{12}}(z_1, z_2; g_3 u, w_1, w_2, w_3') = f^e(z_1, z_2; u, w_1, w_2, w_3'). \tag{1.7}$$

Similarly, on the region $|z_2| > |z_1| > 0$, $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$, the sum of the series $\langle w_3', \mathcal{Y}(w_1, z_2)Y_{W_3}^{g_2}(g_2 u, z_1)w_2\rangle$ is equal to $f^e(z_1, z_2; g_2 u, w_1, w_2, w_3')$. By the equivariance property for $W_2$,

$$\langle w_3', \mathcal{Y}(w_1, z_2)Y_{W_3}^{g_2}(g_2 u, z_1)w_2\rangle = \langle w_3', \mathcal{Y}(w_1, z_2)\left(Y_{W_2}^{g_2}\right)^{-1}(u, z_1)w_2\rangle, \tag{1.8}$$

where $\left(Y_{W_2}^{g_2}\right)^{-1}$ is the $(-1)$-th branch of the twisted vertex operator map $Y_{W_2}^{g_2}$. The right-hand side of (1.8) can be obtained from $\langle w_3', \mathcal{Y}(w_1, z_2)Y_{W_3}^{g_2}(u, z_1)w_2\rangle$ on the region $|z_2| > |z_1| > 0$, $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$ by letting $z_1$ go around clockwise a circle of radius less than $|z_2|$. Such a circle as a loop must have a base point in the region $|z_2| > |z_1| > 0$, $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$. But there is a canonical isomorphism between the fundamental group of $M^2$ with such a base point and $\mathrm{PB}_3$ which has a base point $(-3, -2)$. To see how the loop given by the circle above acts on the single-valued branches of $f(z_1, z_2; u, w_1, w_2, w_3')$, we need to find the element of $\mathrm{PB}_3$ corresponding to this loop. We choose the following loop $\gamma$ based at $(-3, -2)$: The first part $\gamma_1$ of $\gamma$ is the lower half circle centered at $-2$ with radius 1 from $-3$ to $-1$ for $z_1$ and trivial for $z_2$ (meaning $z_2$ is always equal to $-2$). The second part $\gamma_2$ is the counterclockwise circle centered at 0 with radius 1 based at $-1$ for $z_1$ and trivial for $z_2$. The third part $\gamma_3 = \gamma_1^{-1}$ is also the lower half circle centered at $-1$ with radius 1 but from $-1$ to $-3$ for $z_1$ and trivial for $z_2$. It is clear that $\gamma$ is homotopically equivalent to the loop given in Figure 3. Then we have $b_{13} = [\gamma] = [\gamma_1][\gamma_2][\gamma_1]^{-1}$, where $[\gamma]$ for a path $\gamma$ means its homotopy class. Equivalently, we have $[\gamma_2] = [\gamma_1]^{-1} b_{13} [\gamma_1]$. When we let $z_1$ go from $-1$ to $-3$ along $\gamma_1^{-1}$, since $\gamma_1^{-1}$ passes the cut along the positive real line in the $z_1 - z_2$-plane, the single-valued branch $f^e(z_1, z_2; u, w_1, w_2, w_3')$ is changed to the single-valued branch $f^{b_{12}^{-1}}(z_1, z_2; u, w_1, w_2, w_3')$. Similarly, when we let $z_1$ go from

$-1$ to $-3$ along $\gamma_1$, $f^b(z_1, z_2; u, w_1, w_2, w_3')$ is changed to the single-valued branch $f^{bb_{12}}(z_1, z_2; u, w_1, w_2, w_3')$ for any $b \in \mathrm{PB}_3$. Thus when we let $z_1$ go around the loop $\gamma_2$, $f^e(z_1, z_2; u, w_1, w_2, w_3')$ is changed to $f^{b_{12}^{-1}b_{13}b_{12}}(z_1, z_2; u, w_1, w_2, w_3')$, that is,

$$f^{[\gamma_2]}(z_1, z_2; u, w_1, w_2, w_3') = f^{b_{12}^{-1}b_{13}b_{12}}(z_1, z_2; u, w_1, w_2, w_3').$$

Note that the circle $\gamma_2^{-1}$ is exactly the circle we use to obtain the right-hand side of (1.8) from $\langle w_3', \mathcal{Y}(w_1, z_2)Y_{W_3}^{g_2}(u, z_1)w_2\rangle$ on the region $|z_2| > |z_1| > 0$, $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$. So the right-hand side of (1.8) is equal to

$$f^{[\gamma_2^{-1}]}(z_1, z_2; u, w_1, w_2, w_3') = f^{b_{12}^{-1}b_{13}^{-1}b_{12}}(z_1, z_2; u, w_1, w_2, w_3').$$

But the sum of the left-hand side of (1.8) is equal to $f^e(z_1, z_2; g_2u, w_1, w_2, w_3')$. So we obtain

$$f^e(z_1, z_2; g_2u, w_1, w_2, w_3') = f^{b_{12}^{-1}b_{13}^{-1}b_{12}}(z_1, z_2; u, w_1, w_2, w_3')$$

or equivalently

$$f^{b_{12}^{-1}b_{13}b_{12}}(z_1, z_2; g_2u, w_1, w_2, w_3') = f^e(z_1, z_2; u, w_1, w_2, w_3'). \tag{1.9}$$

Similarly we also have

$$f^{b_{12}}(z_1, z_2; g_1u, w_1, w_2, w_3') = f^e(z_1, z_2; u, w_1, w_2, w_3'). \tag{1.10}$$

Using the right action of $\mathrm{PB}_3$ on the set of single-valued branches of the multivalued analytic function $f(z_1, z_2; u, w_1, w_2, w_3')$, for $b \in \mathrm{PB}_3$, we obtain from (1.10)

$$f^{b_{12}b}(z_1, z_2; g_2u, w_1, w_2, w_3') = f^b(z_1, z_2; u, w_1, w_2, w_3'). \tag{1.11}$$

From (1.7), (1.9) and (1.11), we have

$$\begin{aligned} f^{b_{13}b_{12}}(z_1, z_2; g_3u, w_1, w_2, w_3') &= f^e(z_1, z_2; u, w_1, w_2, w_3') = f^{b_{12}^{-1}b_{13}b_{12}}(z_1, z_2; g_2u, w_1, w_2, w_3') \\ &= f^{b_{12}(b_{12}^{-1}b_{13}b_{12})}(z_1, z_2; g_1g_2u, w_1, w_2, w_3') \\ &= f^{b_{13}b_{12}}(z_1, z_2; g_1g_2u, w_1, w_2, w_3') \end{aligned} \tag{1.12}$$

When $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$, the left- and right-hand sides of (1.12) are equal to the sum of $\langle w_3', \left(Y_{W_3}^{g_3}\right)^1(g_3u, z_1)\mathcal{Y}(w_1, z_2)w_2\rangle$ and $\langle w_3', \left(Y_{W_3}^{g_3}\right)^1(g_1g_2u, z_1)\mathcal{Y}(w_1, z_2)w_2\rangle$, respectively. Therefore, we obtain

$$\langle w_3', \left(Y_{W_3}^{g_3}\right)^1(g_1g_2u - g_3u, z_1)\mathcal{Y}(w_1, z_2)w_2\rangle = 0 \tag{1.13}$$

for $w_1 \in W_1$, $w_2 \in W_2$, $w_3 \in W_3'$. Since $\mathcal{Y}$ is surjective in the sense above and $w_3'$ is arbitrary, (1.13) implies $\left(Y_{W_3}^{g_3}\right)^1(g_1g_2u - g_3u, z_1) = 0$. But the map given by $v \to Y_{W_3}^{g_3}(v, x)$ is injective, we obtain $g_1g_2u - g_3u = 0$ for $u \in V$. So we have $g_3 = g_1g_2$. □

## 2. Skew-symmetry and contragredient isomorphisms

Let $g_1$, $g_2$ be automorphisms of $V$, $W_1$, $W_2$ and $W_3$ $g_1$-, $g_2$- and $g_1g_2$-twisted $V$-modules without $g_1$-, $g_2$- and $g_1g_2$-actions, respectively and $\mathcal{Y}$ a twisted intertwining operator of type $\binom{W_3}{W_1\ W_2}$. We define linear maps $\Omega_\pm(\mathcal{Y})\colon W_2 \otimes W_1 \to W_3\{x\}[\log x]$, $w_2 \otimes w_1 \mapsto \Omega_\pm(\mathcal{Y})(w_2, x)w_1$ by

$$\Omega_\pm(\mathcal{Y})(w_2, x)w_1 = \mathrm{e}^{xL(-1)}\mathcal{Y}(w_1, y)w_2\big|_{y^n = \mathrm{e}^{\pm n\pi\mathrm{i}}x^n, \log y = \log x \pm \pi\mathrm{i}} \tag{2.1}$$

for $w_1 \in W_1$ and $w_2 \in W_2$. Note that we can also define $\Omega_p$ for $p \in \mathbb{Z}$ by changing $\pm$ in the right-hand side of (2.1) to $+(2p + 1)$. But we will not discuss $\Omega_p$ in this paper.

From the definition (2.1), for $p \in \mathbb{Z}$, $w_1 \in W_1$, $w_2 \in W_2$ and $z \in \mathbb{C}^\times$,

$$\begin{aligned}\Omega_\pm(\mathcal{Y})^p(w_2,z)w_1 &= \Omega_\pm(\mathcal{Y})(w_2,x)w_1\big|_{x^n=\mathrm{e}^{nl_p(z)},\log x=l_p(z)}\\ &= \left(\mathrm{e}^{xL(-1)}\mathcal{Y}(w_1,y)w_2\big|_{y^n=\mathrm{e}^{\pm n\pi\mathrm{i}}x^n,\log y=\log x\pm\pi\mathrm{i}}\right)\big|_{x^n=\mathrm{e}^{nl_p(z)},\log x=l_p(z)}\\ &= \mathrm{e}^{zL(-1)}\mathcal{Y}(w_1,y)w_2\big|_{y^n=\mathrm{e}^{n(l_p(z)\pm\pi\mathrm{i})},\log y=l_p(z)\pm\pi\mathrm{i}}.\end{aligned}$$

When $\arg z < \pi$ and $\arg z \geq \pi$, $\arg(-z) = \arg z + \pi$ and $\arg(-z) = \arg z - \pi$, respectively. Hence

$$\mathrm{e}^{zL(-1)}\mathcal{Y}(w_1,y)w_2\big|_{y^n=\mathrm{e}^{n(l_p(z)+\pi\mathrm{i})},\log y=l_p(z)+\pi\mathrm{i}} = \mathrm{e}^{zL(-1)}\mathcal{Y}^p(w_1,-z)w_2$$

when $\arg z < \pi$ and

$$\mathrm{e}^{zL(-1)}\mathcal{Y}(w_1,y)w_2\big|_{y^n=\mathrm{e}^{n(l_p(z)-\pi\mathrm{i})},\log y=l_p(z)-\pi\mathrm{i}} = \mathrm{e}^{zL(-1)}\mathcal{Y}^p(w_1,-z)w_2$$

when $\arg z \geq \pi$. In particular, for $w_1 \in W_1$, $w_2 \in W_2$ and $z \in \mathbb{C}^\times$ satisfying $\arg z < \pi$ and $\arg z \geq \pi$, we have

$$\Omega_+(\mathcal{Y})^p(w_2,z)w_1 = \mathrm{e}^{zL(-1)}\mathcal{Y}^p(w_1,-z)w_2 \tag{2.2}$$

and

$$\Omega_-(\mathcal{Y})^p(w_2,z)w_1 = \mathrm{e}^{zL(-1)}\mathcal{Y}^p(w_1,-z)w_2, \tag{2.3}$$

respectively.

Now assume that $W_1$, $W_2$, and $W_3$ are $g_1$-, $g_2$-, and $g_1g_2$-twisted $V$-modules with $g_1$-, $g_2$- and $g_1g_2$-actions, respectively. In the case that $W_1$ also has an action of $g_2^{-1}$, we have the $g_2^{-1}g_1g_2$-twisted $V$-module $\varphi_{g_2^{-1}}(W_1)$. In this case, we define

$$\Omega_+^{g_2^{-1}}(\mathcal{Y})\colon\ W_2 \otimes W_1 \to W_3\{x\}[\log x], \qquad w_2 \otimes w_1 \mapsto \Omega_+^{g_2^{-1}}(\mathcal{Y})(w_2,x)w_1$$

by $\Omega_+^{g_2^{-1}}(\mathcal{Y})(w_2,x)w_1 = \mathrm{e}^{xL(-1)}\mathcal{Y}(g_2w_1,y)w_2\big|_{y^n=\mathrm{e}^{n\pi\mathrm{i}}x^n,\log y=\log x+\pi\mathrm{i}}$ for $w_1 \in W_1$ and $w_2 \in W_2$. In the case that $W_2$ has an action of $g_1$, we have the $g_1g_2g_1^{-1}$-twisted $V$-module $\varphi_{g_1}(W_2)$. In this case, we define

$$\Omega_-^{g_1}(\mathcal{Y})\colon\ W_2 \otimes W_1 \to W_3\{x\}[\log x], \qquad w_2 \otimes w_1 \mapsto \Omega_-^{g_1}(\mathcal{Y})(w_2,x)w_1$$

by $\Omega_-^{g_1}(\mathcal{Y})(w_2,x)w_1 = \mathrm{e}^{xL(-1)}\mathcal{Y}(w_1,y)g_1^{-1}w_2\big|_{y^n=\mathrm{e}^{-n\pi\mathrm{i}}x^n,\log y=\log x-\pi\mathrm{i}}$ for $w_1 \in W_1$ and $w_2 \in W_2$.

For $w_1 \in W_1$, $w_2 \in W_2$ and $z \in \mathbb{C}^\times$ satisfying $\arg z < \pi$ and $\arg z \geq \pi$, we have

$$\Omega_+^{g_2^{-1}}(\mathcal{Y})^p(w_2,z)w_1 = \mathrm{e}^{zL(-1)}g_2\mathcal{Y}^p(g_2w_1,-z)g_2^{-1}w_2 \tag{2.4}$$

and

$$\Omega_-^{g_1}(\mathcal{Y})^p(w_2,z)w_1 = \mathrm{e}^{zL(-1)}\mathcal{Y}^p(w_1,-z)g_1^{-1}w_2, \tag{2.5}$$

respectively.

**Theorem 2.1.** *The linear maps $\Omega_+(\mathcal{Y})$, $\Omega_-(\mathcal{Y})$, $\Omega_+^{g_2^{-1}}(\mathcal{Y})$ and $\Omega_-^{g_1}(\mathcal{Y})$ are twisted intertwining operators of types*

$$\binom{W_3}{W_2\phi_{g_2^{-1}}(W_1)}, \quad \binom{W_3}{\phi_{g_1}(W_2)W_1}, \quad \binom{W_3}{W_2\varphi_{g_2^{-1}}(W_1)}, \quad \binom{W_3}{\varphi_{g_1}(W_2)W_1},$$

*respectively (recall the definition of $\phi_g$ and $\varphi_g$ for an automorphism $g$ of $V$ in Section 1).*

*Proof.* We need only prove the results for $\Omega_+(\mathcal{Y})$ and $\Omega_-(\mathcal{Y})$. Using the definitions of $\varphi_{g_1}$ and $\varphi_{g_2^{-1}}$ and (2.4) and (2.5), we see that the proofs for $\Omega_+^{g_2^{-1}}(\mathcal{Y})$ and $\Omega_-^{g_1}(\mathcal{Y})$ are reduced to the proofs for $\Omega_+(\mathcal{Y})$ and $\Omega_-(\mathcal{Y})$.

The proofs of the lower-truncation property and the $L(-1)$-derivative property of $\Omega_+(\mathcal{Y})$ and $\Omega_-(\mathcal{Y})$ are easy to verify. As in the proof of [17, Theorem 5.1], we

omit their proofs here. The main difference between the proof here and the proof of [17, Theorem 5.1] is that here we cannot use the explicit form of the correlation functions in [17] obtained from the products and iterates of twisted intertwining operators and twisted vertex operators. So our proof here is much more complicated and involves some technical convergence and analytic extension results, even though the idea is the same as in the proof of [17, Theorem 5.1].

Let $u \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w_3' \in W_3'$. As in the proof of [17, Theorem 5.1], we use $f(z_1, z_2; u, w_1, w_2, w_3')$ to denote the multivalued analytic function in the duality property for the twisted intertwining operator $\mathcal{Y}$ with the preferred branch $f^e(z_1, z_2; u, w_1, w_2, w_3')$. Note that $f(z_1, z_2; u, w_1, w_2, w_3')$ in [17] is of the particular form in the definition of twisted intertwining operator there. But in this proof, it is a maximally-extended multivalued analytic function on $M^2$ with regular singular points at $z_1, z_2 = 0$ and $z_1 - z_2 = 0$ and in general might not have the special form in [17].

Define

$$g_{\pm}(z_1, z_2; u, w_2, w_1, w_3') = f\big(z_1 - z_2, -z_2; u, w_1, w_2, \mathrm{e}^{z_2 L'(1)} w_3'\big)$$

and choose the preferred branch $g_{\pm}^e(z_1, z_2; u, w_2, w_1, w_3')$ of $g_{\pm}(z_1, z_2; u, w_2, w_1, w_3')$ as follows: On the subregion $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_2 < \pi$ (for $\Omega_+$) or $\arg z_2 \geq \pi$ (for $\Omega_-$) of $M_0^2$, let

$$g_{\pm}^e(z_1, z_2; u, w_2, w_1, w_3') = f^e\big(z_1 - z_2, -z_2; u, w_1, w_2, \mathrm{e}^{z_2 L'(1)} w_3'\big).$$

For general $(z_1, z_2) \in M_0^2$, we define $g_{\pm}^e(z_1, z_2; u, w_2, w_1, w_3')$ to be the unique analytic extension on $M_0^2$.

When $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_2 < \pi$ (for $\Omega_+$) or $\arg z_2 \geq \pi$ (for $\Omega_-$), from (2.2) and (2.3) and the $L(-1)$-derivative property for $Y_{W_3}^{g_3}$, we have

$$\begin{aligned}
&\big\langle w_3', Y_{W_3}^{g_3}(u, z_1)\Omega_{\pm}(\mathcal{Y})(w_2, z_2)w_1\big\rangle \\
&= \big\langle w_3', Y_{W_3}^{g_3}(u, x_1)\Omega_{\pm}(\mathcal{Y})(w_2, x_2)w_1\big\rangle\big|_{x_1^n = \mathrm{e}^{n\log z_1}, \log x_1 = \log z_1, x_2^n = \mathrm{e}^{n\log z_2}, \log x_2 = \log z_2} \\
&= \big\langle w_3', Y_{W_3}^{g_3}(u, x_1)\mathrm{e}^{-yL(-1)}\mathcal{Y}(w_1, y)w_2\big\rangle\big|_{x_1^n = \mathrm{e}^{n\log z_1}, \log x_1 = \log z_1, y^n = \mathrm{e}^{n\log(-z_2)}, \log y = \log(-z_2)} \\
&= \big\langle \mathrm{e}^{z_2 L'(1)} w_3', (Y_{W_3}^{g_3})(u, x_1 + y) \\
&\qquad \times \mathcal{Y}(w_1, y)w_2\big\rangle\big|_{x_1^n = \mathrm{e}^{n\log z_1}, \log x_1 = \log z_1, y^n = \mathrm{e}^{n\log(-z_2)}, \log y = \log(-z_2)}. \qquad (2.6)
\end{aligned}$$

We first prove that the right-hand side of formula (2.6) is absolutely convergent on the region $|z_1| > |z_2| > 0$ and is convergent to $g_{\pm}^e(z_1, z_2; u, w_1, w_2, w_3')$ on the region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$. The proof is in fact the same as the proof that the right-hand side of [30, equation (9.170)] is absolutely convergent in the region $|z_2| > |z_0| > 0$. Here we give a slightly different proof.

We can always take $u \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $\mathrm{e}^{z_2 L'(1)} w_3' \in W_3'$ to be homogeneous. Let $\Delta = -\mathrm{wt}\, \mathrm{e}^{z_2 L'(1)} w_3' + \mathrm{wt}\, u + \mathrm{wt}\, w_1 + \mathrm{wt}\, w_2$. Let

$$D = \big\{n \in \mathbb{C} \mid \text{there exist } i, j \in \mathbb{N}, \big\langle \big(Y_{W_3}^{g_3}\big)_{\Delta - n - 2, j}(u)\mathcal{Y}_{n,i}(w_1)w_2\big\rangle \neq 0\big\}$$

and $M, N \in \mathbb{N}$ such that $\big(Y_{W_3}^{g_3}\big)_{m,j}(u) = 0$ for $m \in \mathbb{C}$, $j > M$ and $\mathcal{Y}_{n,i}(w_1) = 0$ for $i > N$. Then by the lower truncation property of $\mathcal{Y}$, the fact that $u$ is a finite sum of generalized eigenvectors of $g_3$ and the equivariance property of the $g_3$-twisted module $W_3$, we know that there exist a finite subset $A$ of $\mathbb{C}/\mathbb{Z}$ and $R_\mu \in \mu$ for each $\mu \in A$ such that

$$D \subset \bigcup_{\mu \in A} = (R_\mu - \mathbb{N}).$$

Let

$$a_{n,j,i} = \left\langle \mathrm{e}^{z_2 L'(1)} w_3', \left(Y_{W_3}^{g_3}\right)_{\Delta-n-2,j}(u)(\mathcal{Y})_{n,i}(w_1)w_2 \right\rangle \in \mathbb{C}$$

for $n \in D$, $0 \le j \le M$ and $0 \le n \le N$. Then, by the convergence of (1.3), the $L(-1)$-derivative properties for $Y_{W_3}^{g_3}$ and $\mathcal{Y}$, and [29, Proposition 7.9], we know that the triple series

$$\sum_{n\in D}\sum_{j=0}^{M}\sum_{i=0}^{N} a_{n,j,i}\mathrm{e}^{(-\Delta+n+1)\log z_1}(\log z_1)^j \mathrm{e}^{(-n-1)\log z_2}(\log z_2)^i \tag{2.7}$$

is absolutely convergent on the region given by $|z_1| > |z_2| > 0$ and is convergent on the region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ to

$$f^e\left(z_1, z_2; u, w_1, w_2, \mathrm{e}^{z_2 L'(1)} w_3'\right) = \left\langle \mathrm{e}^{z_2 L'(1)} w_3', Y_{W_3}^{g_3}(u, z_1)\mathcal{Y}(w_1, z_2)w_2 \right\rangle.$$

For $n \in D$, $j = 0, \dots, M$, $k \in \mathbb{Z}_{\ge 0}$, $s = 0, \dots, j$, let $b_{n,j,k,s} \in \mathbb{C}$ be the numbers defined in (3.4). Then

$$\sum_{m\in D-\mathbb{N}}\sum_{s=0}^{M}\sum_{i=0}^{N}\left(\sum_{j=s}^{M}\sum_{\substack{n-k=m\\ n\in D,\, k\in\mathbb{N}}} a_{n,j,i}b_{n,j,k,s}\right)\mathrm{e}^{(-\Delta+m+1)\log z_1}(\log z_1)^s \times \mathrm{e}^{(-m-1)\log(-z_2)}(\log(-z_2))^i$$

is equal to the right-hand side of (2.6) and, by Lemma 3.1, is absolutely convergent on the region $|z_1| > |z_2| > 0$ and is convergent to $f^e\left(z_1 - z_2, -z_2; u, w_1, w_2, \mathrm{e}^{z_2 L'(1)} w_3'\right)$ on the region $|z_1| > |z_2| > 0$, $|\arg z_1 - \arg(z_1 - z_2)| < \frac{\pi}{2}$.

Now it is easy to see that the left-hand side of (2.6) is absolutely convergent on the region $|z_1| > |z_2| > 0$ and its sum is equal to $g_{\pm}^e(z_1, z_2; u, w_1, w_2, w_3')$ on the region $|z_1| > |z_2| > 0$, $|\arg z_1 - \arg(z_1 - z_2)| < \frac{\pi}{2}$. In fact, we know that the left-hand side of (2.6) as a series is equal to the right-hand side of (2.6) on the region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_2 < \pi$ (for $\Omega_+$) or $\arg z_2 \ge \pi$ (for $\Omega_-$). But we have just proved that the right-hand side of (2.6) is absolutely convergent on the larger region $|z_1| > |z_2| > 0$. The left-hand side of (2.6) is also a series of the same form as (2.7). In particular, its absolute convergence on the smaller region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_2 < \pi$ (for $\Omega_+$) or $\arg z_2 \ge \pi$ (for $\Omega_-$) implies its absolute convergence on the larger region $|z_1| > |z_2| > 0$.

On the region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_2 < \pi$ (for $\Omega_+$) or $\arg z_2 \ge \pi$ (for $\Omega_-$), by (2.6) and the discussion above, the left-hand side of (2.6) is convergent to $f^e\left(z_1 - z_2, -z_2; u, w_1, w_2, \mathrm{e}^{z_2 L'(1)} w_3'\right)$, which in turn is by definition equal to $g_{\pm}^e(z_1, z_2; u, w_1, w_2, w_3')$ on the same region. We know that the left-hand side of (2.6) on the region given by $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ with cuts along the positive real lines on the $z_1$- and $z_2$-planes is convergent to the analytic extension of the sum of the left-hand side of (2.6) on the smaller region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_2 < \pi$ (for $\Omega_+$) or $\arg z_2 \ge \pi$ (for $\Omega_-$). Also, by definition, $g_{\pm}^e(z_1, z_2; u, w_1, w_2, w_3')$ on $M_0^2$ is obtained by analytically extending $g_{\pm}^e(z_1, z_2; u, w_1, w_2, w_3')$ on the smaller region $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_2 < \pi$ (for $\Omega_+$) or $\arg z_2 \ge \pi$ (for $\Omega_-$). Thus the left-hand side of (2.6) on the region given by $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ is absolutely convergent to $g_{\pm}^e(z_1, z_2; u, w_1, w_2, w_3')$ on $M_0^2$.

Generalizing the proof of the convergence of (2.6) above, we can prove that

$$\left\langle w_3', Y_{W_3}^{g_3}(u_1, z_1)\cdots Y_{W_3}^{g_3}(u_{k-1}, z_{k-1})\Omega_{\pm}(\mathcal{Y})(w_2, z_k)w_1 \right\rangle \tag{2.8}$$

is absolutely convergent on the region $|z_1| > \cdots > |z_k| > 0$ and can be maximally extended to a multivalued analytic function on the region $M^k$ for $k \in \mathbb{N} + 3$, $u_1, \dots, u_{k-1} \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w_3' \in W_3'$. In fact, generalizing (2.6), we see that (2.8) is equal to

$$\begin{aligned} &\left\langle \mathrm{e}^{z_k L'(1)} w_3', \left(Y_{W_3}^{g_3}\right)(u_1, x_1 + y) \cdots \left(Y_{W_3}^{g_3}\right)(u_{k-1}, x_{k-1} + y) \right. \\ &\left. \times \mathcal{Y}(w_1, y) w_2 \right\rangle \Big|_{x_i^n = \mathrm{e}^{n \log z_i}, \log x_i = \log z_i, i = 1, \dots, k-1, y^n = \mathrm{e}^{n \log(-z_k)}, \log y = \log(-z_k)}. \end{aligned} \tag{2.9}$$

From Definition 1.1, we see that the convergence and analytic extension of (2.9) on the region $|z_1| > \cdots > |z_k| > 0$ is equivalent to the convergence and analytic extension of

$$\begin{aligned} &\prod_{1 \le i < j \le k-1} (z_i - z_j)^{M_{ij}} \left\langle \mathrm{e}^{z_k L'(1)} w_3', \left(Y_{W_3}^{g_3}\right)(u_1, x_1 + y) \cdots \left(Y_{W_3}^{g_3}\right)(u_{k-1}, x_{k-1} + y) \right. \\ &\left. \times \mathcal{Y}(w_1, y) w_2 \right\rangle \Big|_{x_i^n = \mathrm{e}^{n \log z_i}, \log x_i = \log z_i, i = 1, \dots, k-1, y^n = \mathrm{e}^{n \log(-z_k)}, \log y = \log(-z_k)} \end{aligned} \tag{2.10}$$

on the region $|z_i| > |z_k| > 0$ for $i = 1, \dots, k-1$, $z_i \ne z_j$ for $i \ne j$, where $M_{ij} \in \mathbb{Z}_+$ for $i \ne j$ satisfy $x^{M_{ij}} Y_V(u_i, x) u_j \in V[[x]]$. From Remark 3.3, we know that Lemma 3.1 can be generalized to the case of more than two variables for a series of the form (2.10). It is important to note that the polynomial $\prod_{1 \le i < j \le k-1} (z_i - z_j)^{M_{ij}}$ in (2.10) is multiplied to cancel the poles at $z_i = z_j$ for $1 \le i < j \le k-1$ so that the series (2.10) is a different iterated sum of a convergent iterated series of the form (3.22). Using the convergence of products with more than one twisted vertex operator for the twisted intertwining operator $\mathcal{Y}$ and this generalization of Lemma 3.1, we see that (2.10) is absolutely convergent on the region $|z_i| > |z_k| > 0$ for $i = 1, \dots, k-1$, $z_i \ne z_j$ for $i \ne j$ and its sum has analytic extension on the region $M^k$. Thus (2.8) is absolutely convergent on the region $|z_1| > \cdots > |z_k| > 0$ and its sum has maximal analytic extension on the region $M^k$. Moreover, from the discussion above, we have

$$g(z_1, \dots, z_k) = \sum_{j=0}^{J} \frac{z_k^j}{j!} f_j(z_1 - z_k, \dots, z_{k-1} - z_k, -z_k),$$

where $g(z_1, \dots, z_k)$ is the maximal analytic extension of the sum of (2.8) on the region $M^k$, $J \in \mathbb{Z}_+$ such that $L'(1)^{J+1} w_3' = 0$, and $f_j(z_1, \dots, z_k)$ for $j = 0, \dots, J$ are the maximal analytic extensions of the sum of

$$\left\langle L'(1)^j w_3', Y_{W_3}^{g_3}(u_1, z_1) \cdots Y_{W_3}^{g_3}(u_{k-1}, z_{k-1}) \mathcal{Y}(w_1, z_k) w_2 \right\rangle.$$

For fixed $z_k \in \mathbb{C}^\times$, let $(z_1, \dots, z_{k-1}) - \beta = \delta$ for $\beta = \mathbb{C}^{k-1}$ and $\delta \in \{0, \infty\}^{k-1}$ be a component-isolated singularity of the multivalued analytic function of of $z_1, \dots, z_{k-1}$ obtained as a value $\prod_{1 \le i < j \le k-1} (z_i - z_j)^{M_{ij}} g(z_1, \dots, -z_k)$ at $\zeta = -z_k$ of $\prod_{1 \le i < j \le k-1} (z_i - z_j)^{M_{ij}} g(z_1, \dots, \zeta)$. Then this singularity, which can also be written as $(z_1 - z_k, \dots, z_{k-1} - z_k) - (\beta + (z_k, \dots, z_k)) = \delta$, is also a component-isolated singularity of the corresponding values

$$\prod_{1 \le i < j \le k-1} (z_i - z_j)^{M_{ij}} f_j(z_1 - z_k, \dots, z_{k-1} - z_k, -z_k)$$

at $\zeta = -z_k$ of

$$\prod_{1 \le i < j \le k-1} (z_i - z_j)^{M_{ij}} f_j(z_1 - z_k, \dots, z_{k-1} - z_k, \zeta)$$

for $j = 0, \dots, J$. Thus this singularity of

$$\prod_{1 \le i < j \le k-1} (z_i - z_j)^{M_{ij}} f_j(z_1 - z_k, \dots, z_{k-1} - z_k, -z_k)$$

corresponds to the component-isolated singularity $(z_1, \dots, z_{k-1}) - (\beta + (z_k, \dots, z_k)) = \delta$ of $\prod_{1 \le i < j \le k-1} (z_i - z_j)^{M_{ij}} f_j(z_1, \dots, z_k)$. But we know that a component-isolated singularity of $\prod_{1 \le i < j \le k-1} (z_i - z_j)^{M_{ij}} f_j(z_1, \dots, z_{k-1}, -z_k)$ of such a form is regular. So this component-isolated singularity $(z_1, \dots, z_{k-1}) - (\beta + (z_k, \dots, z_k)) = \delta$ is regular. Equivalently, the component-isolated singularity $(z_1, \dots, z_{k-1}) - \beta = \delta$ of

$$\prod_{1 \le i < j \le k-1} (z_i - z_j)^{M_{ij}} f_j(z_1 - z_k, \dots, z_{k-1} - z_k, -z_k)$$

for $j = 0, \dots, J$ is regular and thus the component-isolated singularity of $\prod_{1 \le i < j \le k-1} (z_i - z_j)^{M_{ij}} g(z_1, \dots, z_{k-1}, -z_k)$ is regular. But $z_k$ is arbitrary, $-z_k$ is also arbitrary. We see that for any fixed $z_k \in \mathbb{C}^\times$, $(z_1, \dots, z_{k-1}) - \beta = \delta$ is a regular singularity of any value at $\zeta = z_k$ of $\prod_{1 \le i < j \le k-1} (z_i - z_j)^{M_{ij}} g(z_1, \dots, \zeta)$.

When $|z_2| > |z_1| > 0$ and $\arg z_2 \ge \pi$,

$$\begin{aligned} &\big\langle w_3', \Omega_-(\mathcal{Y})(w_2, z_2) Y_{W_1}^{g_1}(u, z_1) w_1 \big\rangle \\ &= \big\langle w_3', \mathrm{e}^{z_2 L(-1)} \mathcal{Y}\big(Y_{W_1}^{g_1}(u, z_1) w_1, -z_2\big) w_2 \big\rangle \\ &= \big\langle \mathrm{e}^{z_2 L'(1)} w_3', \mathcal{Y}\big(Y_{W_1}^{g_1}(u, (z_1 - z_2) - (-z_2)) w_1, -z_2\big) w_2 \big\rangle \end{aligned} \tag{2.11}$$

converges absolutely and if in addition, $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$, its sum is equal to

$$f^e\big(z_1 - z_2, -z_2; u, w_1, w_2, \mathrm{e}^{z_2 L'(1)} w_3'\big). \tag{2.12}$$

Note that by definition, (2.12) is a single-valued analytic function on the set $\widetilde{M_0^2}$ given by cutting $M^2$ along the positive real line in the $z_1$- and $(z_1 - z_2)$-planes and along the negative real line in the $z_2$-plane, with these positive real lines in the $z_1$- and $(z_1 - z_2)$-planes attached to the upper half $z_1$- and $(z_1 - z_2)$-planes and the negative real line in the $z_2$-plane attached to the lower half $z_2$-plane. Then the subset

$$\left\{(z_1, z_2) \in M_0^2 \mid |z_2| > |z_1| > 0,\ \arg z_2 \ge \pi,\ |\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}\right\} \tag{2.13}$$

of $\widetilde{M_0^2}$ is also a subset of $M_0^2$. But on the subset of $M_0^2$ given by $|z_1| > |z_2| > 0$, $\arg z_2 \ge \pi$, by definition, (2.12) is equal to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$. Since $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ on $M_0^2$ is obtained by analytic extension, we see that (2.12) is equal to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ also on the subset (2.13). Thus the left-hand side of (2.11) is absolutely convergent to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_2| > |z_1| > 0$, $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$ and $\arg z_2 \ge \pi$. Since $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ and the sum of (2.11) are both analytic extensions of their restrictions on the subset given by $|z_2| > |z_1| > 0$, $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$ and $\arg z_2 \ge \pi$, we see that the left-hand side of (2.11) is absolutely convergent to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_2| > |z_1| > 0$ and $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$. But when $\arg z_2 \ge \pi$, $\arg(-z_2) = \arg z_2 - \pi$. Hence in this case, the inequality $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$ becomes $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$. Also both the left-hand side of (2.11) and $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ are single valued analytic functions in $z_1$ and $z_2$ with cuts at $z_1 \in \mathbb{R}_+$ and $z_2 \in \mathbb{R}_+$. Thus when $|z_2| > |z_1| > 0$ and $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$, the left-hand side of (2.11) is absolutely convergent to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$.

Next we discuss the iterate of $\Omega_-(\mathcal{Y})$ and the twisted vertex operator map $\phi_{g_1}\big(Y_{W_2}^{g_2}\big)$. When $|z_2| > |z_1 - z_2| > 0$ and $\arg z_2 \geq \pi$,

$$\begin{aligned}
&\big\langle w_3', \Omega_-(\mathcal{Y})\big(\phi_{g_1}\big(Y_{W_2}^{g_2}\big)(u, z_1 - z_2)w_2, z_2\big)w_1\big\rangle \\
&= \big\langle w_3', \Omega_-(\mathcal{Y})\big(\big(Y_{W_2}^{g_2}\big)\big(g_1^{-1}u, z_1 - z_2\big)w_2, z_2\big)w_1\big\rangle \\
&= \big\langle w_3', \mathrm{e}^{z_2 L(-1)}\mathcal{Y}(w_1, -z_2)\big(Y_{W_2}^{g_2}\big)\big(g_1^{-1}u, z_1 - z_2\big)w_2\big\rangle \\
&= \big\langle \mathrm{e}^{z_2 L'(1)}w_3', \mathcal{Y}(w_1, -z_2)\big(Y_{W_2}^{g_2}\big)\big(g_1^{-1}u, z_1 - z_2\big)w_2\big\rangle \qquad (2.14)
\end{aligned}$$

converges absolutely and if in addition, $-\frac{3\pi}{2} < \arg z_1 - \arg(-z_2) < -\frac{\pi}{2}$, its sum is equal to $f^e(z_1 - z_2, -z_2; g_1^{-1}u, w_1, w_2, \mathrm{e}^{z_2 L'(1)}w_3')$. Note that the proofs of [17, Lemmas 4.5 and 4.6] do not use the explicit form of the multivalued analytic functions in the duality property of the twisted intertwining operators introduced in [17]. Then the same proof of [17, Lemma 4.5] shows that the sum of the right-hand side of (2.14) is equal to $f^{b_{12}^{-1}}\big(z_1 - z_2, -z_2; g_1^{-1}u, w_1, w_2, \mathrm{e}^{z_2 L'(1)}w_3'\big)$ when $|z_2| > |z_1 - z_2| > 0$, $\arg z_2 \geq \pi$ and $\frac{\pi}{2} < \arg z_1 - \arg(-z_2) < \frac{3\pi}{2}$. The same proof of (4.4) in [17, Lemma 4.6] shows that

$$f^{b_{12}^{-1}}\big(z_1 - z_2, -z_2; g_1^{-1}u, w_1, w_2, \mathrm{e}^{z_2 L'(1)}w_3'\big) = f^e\big(z_1 - z_2, -z_2; u, w_1, w_2, \mathrm{e}^{z_2 L'(1)}w_3'\big).$$

Since when $\arg z_2 \geq \pi$, $\arg(-z_2) = \arg z_2 - \pi$ and $\frac{\pi}{2} < \arg z_1 - \arg(-z_2) < \frac{3\pi}{2}$ becomes $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$, we see that the sum of the right-hand side of (2.14) is equal to (2.12) when $|z_2| > |z_1 - z_2| > 0$, $\arg z_2 \geq \pi$ and $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$.

We now use the same argument as above to finish the proof in this case. The subset

$$\Big\{(z_1, z_2) \in M_0^2 \mid |z_2| > |z_1 - z_2| > 0,\ \arg z_2 \geq \pi,\ |\arg(z_1) - \arg z_2| < \frac{\pi}{2}\Big\} \qquad (2.15)$$

of $\widetilde{M}_0^2$ is also a subset of $M_0^2$. By definition, on the subset of $M_0^2$ given by $|z_1| > |z_2| > 0$, $\arg z_2 \geq \pi$, (2.12) is equal to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$. Since $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ on $M_0^2$ is obtained by analytic extension, we see that (2.12) is equal to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ also on the subset (2.15). Thus the left-hand side of (2.14) is absolutely convergent to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_2| > |z_1 - z_2| > 0$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$ and $\arg z_2 \geq \pi$. Since $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ and the sum of (2.14) are both analytic extensions of their restrictions on the subset given by $|z_2| > |z_1 - z_2| > 0$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$ and $\arg z_2 \geq \pi$, we see that the left-hand side of (2.14) is absolutely convergent to $g_-^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_2| > |z_1 - z_2| > 0$ and $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$.

We still need to prove the two other cases for $\Omega_+(\mathcal{Y})$. When $|z_2| > |z_1| > 0$ and $\arg z_2 < \pi$,

$$\begin{aligned}
&\big\langle w_3', \Omega_+(\mathcal{Y})(w_2, z_2)\phi_{g_2^{-1}}\big(Y_{W_1}^{g_1}\big)(u, z_1)w_1\big\rangle \\
&= \big\langle w_3', \mathrm{e}^{z_2 L(-1)}\mathcal{Y}\big(\phi_{g_2^{-1}}\big(Y_{W_1}^{g_1}\big)(u, z_1)w_1, -z_2\big)w_2\big\rangle \\
&= \big\langle \mathrm{e}^{z_2 L'(1)}w_3', \mathcal{Y}\big(\big(Y_{W_1}^{g_1}\big)(g_2 u, z_1)w_1, -z_2\big)w_2\big\rangle \qquad (2.16)
\end{aligned}$$

converges absolutely and if in addition, $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$, its sum is equal to

$$f^e\big(z_1 - z_2, -z_2; g_2 u, w_1, w_2, \mathrm{e}^{z_2 L'(1)}w_3'\big). \qquad (2.17)$$

The subset

$$\Big\{(z_1, z_2) \in M_0^2 \mid |z_2| > |z_1| > 0,\ \arg z_2 < \pi,\ |\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}\Big\} \qquad (2.18)$$

of $\widetilde{M}_0^2$ is also a subset of $M_0^2$. By definition, on the subset of $M_0^2$ given by $|z_1| > |z_2| > 0$, $\arg z_2 < \pi$, (2.17) is equal to $g_+^e(z_1, z_2; g_2 u, w_2, w_1, w_3')$. Since

$g_+^e(z_1, z_2; g_2u, w_2, w_1, w_3')$ on $M_0^2$ is obtained by analytic extension, (2.17) is equal to $g_+^e(z_1, z_2; g_2u, w_2, w_1, w_3')$ also on the subset (2.18). Thus the left-hand side of (2.16) is absolutely convergent to $g_+^e(z_1, z_2; g_2u, w_2, w_1, w_3')$ when $|z_2| > |z_1| > 0$, $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$ and $\arg z_2 < \pi$. Since $g_+^e(z_1, z_2; g_2u, w_2, w_1, w_3')$ and the sum of (2.16) are both analytic extensions of their restrictions on the subset given by $|z_2| > |z_1| > 0$, $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$ and $\arg z_2 < \pi$, the left-hand side of (2.16) is absolutely convergent to $g_+^e(z_1, z_2; g_2u, w_2, w_1, w_3')$ when $|z_2| > |z_1| > 0$, $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$.

The same proof as that of (4.5) in [17, Lemma 4.6] gives $g_+^e(z_1, z_2; g_2u, w_1, w_2, w_3') = g_+^{b_{13}}(z_1, z_2; u, w_1, w_2, w_3')$, since $\arg z_1 < \pi$, $\arg(-z_2) = \arg z_2 + \pi$ and the inequality $|\arg(z_1 - z_2) - \arg(-z_2)| < \frac{\pi}{2}$ becomes $\frac{\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < \frac{3\pi}{2}$. So the sum of the left-hand side of (2.16) is absolutely convergent to $g_+^{b_{13}}(z_1, z_2; u, w_1, w_2, w_3')$ when $|z_2| > |z_1| > 0$ and $\frac{\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < \frac{3\pi}{2}$. Then the same proof as that of [17, Lemma 4.5] shows that this is equivalent to the sum of left-hand side of (2.16) being equal to $g_+^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_2| > |z_1| > 0$ and $-\frac{3\pi}{2} < \arg(z_1 - z_2) - \arg z_2 < -\frac{\pi}{2}$.

Finally, we discuss the iterate of $\Omega_+(\mathcal{Y})$ and the twisted vertex operator map $Y_{W_2}^{g_2}$. When $|z_2| > |z_1 - z_2| > 0$ and $\arg z_2 < \pi$,

$$\begin{aligned} &\big\langle w_3', \Omega_+(\mathcal{Y})\big(Y_{W_2}^{g_2}(u, z_1 - z_2)w_2, z_2\big)w_1\big\rangle \\ &= \big\langle w_3', \mathrm{e}^{z_2 L(-1)}\mathcal{Y}(w_1, -z_2)Y_{W_2}^{g_2}(u, z_1 - z_2)w_2\big\rangle \\ &= \big\langle \mathrm{e}^{z_2 L'(1)}w_3', \mathcal{Y}(w_1, -z_2)Y_{W_2}^{g_2}(u, z_1 - z_2)w_2\big\rangle \end{aligned} \tag{2.19}$$

converges absolutely and if in addition, $-\frac{3\pi}{2} < \arg z_1 - \arg(-z_2) < -\frac{\pi}{2}$, its sum is equal to (2.12). The subset

(2.20)
$$\left\{(z_1, z_2) \in M_0^2 \mid |z_2| > |z_1 - z_2| > 0,\ \arg z_2 < \pi,\ -\frac{3\pi}{2} < \arg z_1 - \arg(-z_2) < -\frac{\pi}{2}\right\}$$

of $\widetilde{M_0^2}$ is also a subset of $M_0^2$. As we have discussed above, on the subset of $M_0^2$ given by $|z_1| > |z_2| > 0$, $\arg z_2 < \pi$, (2.12) is equal to $g_+^e(z_1, z_2; u, w_2, w_1, w_3')$. Since $g_+^e(z_1, z_2; u, w_2, w_1, w_3')$ on $M_0^2$ is obtained by analytic extension, (2.12) is equal to $g_+^e(z_1, z_2; u, w_2, w_1, w_3')$ also on the subset (2.20). Thus the left-hand side of (2.19) is absolutely convergent to $g_+^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_2| > |z_1 - z_2| > 0$, $-\frac{3\pi}{2} < \arg z_1 - \arg(-z_2) < -\frac{\pi}{2}$ and $\arg z_2 < \pi$. When $\arg z_2 < \pi$, $\arg(-z_2) = \pi + \arg z_2$. So in this case, $-\frac{3\pi}{2} < \arg z_1 - \arg(-z_2) < -\frac{\pi}{2}$ is the same as $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$. Since $g_+^e(z_1, z_2; u, w_2, w_1, w_3')$ and the sum of (2.19) are both analytic extensions of their restrictions on the subset given by $|z_2| > |z_1 - z_2| > 0$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$ and $\arg z_2 < \pi$, the left-hand side of (2.19) is absolutely convergent to $g_+^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_2| > |z_1 - z_2| > 0$ and $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$. □

Let $\mathcal{V}_{W_1W_2}^{W_3}$ be the space of twisted intertwining operators of type $\binom{W_3}{W_1W_2}$. Then we have the following.

**Corollary 2.2.** *The maps*

$$\Omega_+ \colon\ \mathcal{V}_{W_1W_2}^{W_3} \to \mathcal{V}_{W_2\phi_{g_2^{-1}}(W_1)}^{W_3}, \qquad \Omega_- \colon\ \mathcal{V}_{W_1W_2}^{W_3} \to \mathcal{V}_{\phi_{g_1}(W_2)W_1}^{W_3},$$

$$\Omega_+^{g_2^{-1}} \colon\ \mathcal{V}_{W_1W_2}^{W_3} \to \mathcal{V}_{W_2\varphi_{g_2^{-1}}(W_1)}^{W_3}, \qquad \Omega_-^{g_1} \colon\ \mathcal{V}_{W_1W_2}^{W_3} \to \mathcal{V}_{\varphi_{g_1}(W_2)W_1}^{W_3}$$

*are linear isomorphisms. In particular,*

$$\mathcal{V}_{W_1W_2}^{W_3}, \quad \mathcal{V}_{\phi_{g_1}(W_2)W_1}^{W_3}, \quad \mathcal{V}_{W_2\phi_{g_2^{-1}}(W_1)}^{W_3}, \quad \mathcal{V}_{\varphi_{g_1}(W_2)W_1}^{W_3} \quad \textit{and} \quad \mathcal{V}_{W_2\varphi_{g_2^{-1}}(W_1)}^{W_3}$$

*are linearly isomorphic.*

*Proof.* It is clear that $\Omega_+$ and $\Omega_-$ are inverse of each other. Since ${}_1\varphi_{g_1}(W_2)$ and $\varphi_{g_2^{-1}}(W_1)$ are equivalent to $\phi_{g_1}(W_2)$ and $\phi_{g_2^{-1}}(W_1)$, respectively, $\Omega_+^{g_2}$ and $\Omega_-^{g_1}$ are also linear isomorphisms. □

The linear isomorphisms $\Omega_+$, $\Omega_-$, $\Omega_+^{g_2^{-1}}$ and $\Omega_-^{g_1}$ are called *skew-symmetry isomorphisms*.

Let $g_1$, $g_2$ be automorphisms of $V$, $W_1$, $W_2$ and $W_3$ $g_1$-, $g_2$- and $g_1g_2$-twisted $V$-modules without $g_1$-, $g_2$- and $g_1g_2$-actions and $\mathcal{Y}$ a twisted intertwining operator of type $\binom{W_3}{W_1\,W_2}$. We define linear maps $A_\pm(\mathcal{Y})\colon W_1\otimes W_3'\to W_2'\{x\}[\log x]$, $w_1\otimes w_3'\mapsto A_\pm(\mathcal{Y})(w_1,x)w_3'$ by

$$\langle A_\pm(\mathcal{Y})(w_1,x)w_3',w_2\rangle=\big\langle w_3',\mathcal{Y}\big(\mathrm{e}^{xL(1)}\mathrm{e}^{\pm\pi\mathrm{i}L(0)}\big(x^{-L(0)}\big)^2w_1,x^{-1}\big)w_2\big\rangle \tag{2.21}$$

for $w_1\in W_1$ and $w_2\in W_2$ and $w_3'\in W_3'$. Similarly to $\Omega_p$ for $p\in\mathbb{Z}$, we can also define $A_p$ for $p\in\mathbb{Z}$ by replacing $\pm$ in (2.21) by $(2p+1)$. But we will also not discuss $A_p$ in this paper.

Let $L_{W_1}^s(0)$ be the semisimple part of $L_{W_1}(0)$. From the definition (2.21), for $p\in\mathbb{Z}$, $w_1\in W_1$, $w_2\in W_2$, $w_3'\in W_3'$ and $z\in\mathbb{C}^\times$, we have

$$\begin{aligned}
&\big\langle A_\pm(\mathcal{Y})^p(w_1,z)w_3',w_2\big\rangle\\
&=\big\langle A_\pm(\mathcal{Y})^p(w_1,x)w_3',w_2\big\rangle\big|_{x^n=\mathrm{e}^{nl_p(z)},\log x=l_p(z)}\\
&=\big\langle w_3',\mathcal{Y}\big(\mathrm{e}^{xL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\big(x^{-L_{W_1}(0)}\big)^2w_1,x^{-1}\big)w_2\big\rangle\big|_{x^n=\mathrm{e}^{nl_p(z)},\log x=l_p(z)}\\
&=\big\langle w_3',\mathcal{Y}\big(\mathrm{e}^{xL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\big(x^{-L_{W_1}^s(0)}\big)^2\cdot\\
&\quad\times\big(\mathrm{e}^{-(L_{W_1}(0)-L_{W_1}^s(0))\log x}\big)^2w_1,x^{-1}\big)w_2\big\rangle\big|_{x^n=\mathrm{e}^{nl_p(z)},\log x=l_p(z)}\\
&=\big\langle w_3',\mathcal{Y}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\big(\mathrm{e}^{-l_p(z)L_{W_1}^s(0)}\big)^2\cdot\\
&\quad\times\big(\mathrm{e}^{-(L_{W_1}(0)-L_{W_1}^s(0))l_p(z)}\big)^2w_1,y\big)w_2\big\rangle\big|_{y^n=\mathrm{e}^{-nl_p(z)},\log y=-l_p(z)}\\
&=\big\langle w_3',\mathcal{Y}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2l_p(z)L_{W_1}(0)}w_1,y\big)w_2\big\rangle\big|_{y^n=\mathrm{e}^{-nl_p(z)},\log y=-l_p(z)}.
\end{aligned}\tag{2.22}$$

When $\arg z=0$, $\arg z^{-1}=\arg z=0$ and $-l_p(z)=l_{-p}(z^{-1})$. When $\arg z\neq 0$, $\arg z^{-1}=-\arg z+2\pi$ and $-l_p(z)=l_{-p-1}(z^{-1})$. Hence when $\arg z=0$, the right-hand side of (2.22) is equal to

$$\begin{aligned}
&\big\langle w_3',\mathcal{Y}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p}(z^{-1})L_{W_1}(0)}w_1,y\big)w_2\big\rangle\big|_{y^n=\mathrm{e}^{nl_{-p}(z^{-1})},\log y=l_{-p}(z^{-1})}\\
&=\big\langle w_3',\mathcal{Y}^{-p}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p}(z^{-1})L_{W_1}(0)}w_1,z^{-1}\big)w_2\big\rangle
\end{aligned}\tag{2.23}$$

and when $\arg z\neq 0$, it is equal to

$$\begin{aligned}
&\big\langle w_3',\mathcal{Y}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p-1}(z^{-1})L_{W_1}(0)}w_1,y\big)w_2\big\rangle\big|_{y^n=\mathrm{e}^{nl_{-p-1}(z^{-1})},\log y=l_{-p-1}(z^{-1})}\\
&=\big\langle w_3',\mathcal{Y}^{-p-1}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p-1}(z^{-1})L_{W_1}(0)}w_1,z^{-1}\big)w_2\big\rangle.
\end{aligned}\tag{2.24}$$

From (2.22)–(2.24), for $w_1\in W_1$, $w_2\in W_2$, $w_3'\in W_3'$ and $z\in\mathbb{C}^\times$, we have

$$\big\langle A_\pm(\mathcal{Y})^p(w_1,z)w_3',w_2\big\rangle=\big\langle w_3',\mathcal{Y}^{-p}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p}(z^{-1})L_{W_1}(0)}w_1,z^{-1}\big)w_2\big\rangle \tag{2.25}$$

when $\arg z=0$ and

$$\begin{aligned}
&\big\langle A_\pm(\mathcal{Y})^p(w_1,z)w_3',w_2\big\rangle\\
&=\big\langle w_3',\mathcal{Y}^{-p-1}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pm\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p-1}(z^{-1})L_{W_1}(0)}w_1,z^{-1}\big)w_2\big\rangle
\end{aligned}\tag{2.26}$$

when $\arg z\neq 0$.

Now assume that $W_1$, $W_2$, and $W_3$ are $g_1$-, $g_2$- and $g_1g_2$-twisted $V$-modules with $g_1$-, $g_2$- and $g_1g_2$-actions, respectively. In the case that $W_2'$ has an action of $g_1$ (which induces a $g_1^{-1}$-action on $W_2$), we have the $g_1g_2^{-1}g_1^{-1}$-twisted $V$-module $\varphi_{g_1}(W_2')$. In this case, we define

$$A_+^{g_1}(\mathcal{Y})\colon\ W_1\otimes W_3'\to W_2'\{x\}[\log x],\qquad w_1\otimes w_3'\mapsto A_+^{g_1}(\mathcal{Y})(w_1,x)w_3'$$

by

$$\big\langle A_+^{g_1}(\mathcal{Y})(w_1,x)w_3',w_2\big\rangle=\big\langle w_3',\mathcal{Y}\big(\mathrm{e}^{xL(1)}\mathrm{e}^{\pi\mathrm{i}L(0)}\big(x^{-L(0)}\big)^2w_1,x^{-1}\big)g_1^{-1}w_2\big\rangle$$

for $w_1\in W_1$, $w_2\in W_2$ and $w_3'\in W_3'$. In the case that $W_3'$ has an action of $g_1^{-1}$ (which induces a $g_1$-action on $W_3$), we have the $g_1^{-1}g_2^{-1}$-twisted $V$-module $\varphi_{g_1^{-1}}(W_3')$. In this case, we define

$$A_-^{g_1^{-1}}(\mathcal{Y})\colon\ W_1\otimes W_3'\to W_2'\{x\}[\log x],\qquad w_1\otimes w_3'\mapsto A_-^{g_1^{-1}}(\mathcal{Y})(w_1,x)w_3'$$

by

$$\big\langle A_-^{g_1^{-1}}(\mathcal{Y})(w_1,x)w_3',w_2\big\rangle=\big\langle g_1w_3',\mathcal{Y}\big(\mathrm{e}^{xL(1)}\mathrm{e}^{-\pi\mathrm{i}L(0)}\big(x^{-L(0)}\big)^2w_1,x^{-1}\big)w_2\big\rangle$$

for $w_1\in W_1$, $w_2\in W_2$ and $w_3'\in W_3'$.

For $w_1\in W_1$, $w_2\in W_2$, $w_3'\in W_3'$ and $z\in\mathbb{C}^\times$, we have from (2.25) and (2.26)

$$\begin{aligned}&\big\langle A_+^{g_1}(\mathcal{Y})^p(w_1,z)w_3',w_2\big\rangle\\&=\big\langle w_3',\mathcal{Y}^{-p}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p}(z^{-1})L_{W_1}(0)}w_1,z^{-1}\big)g_1^{-1}w_2\big\rangle\end{aligned}\tag{2.27}$$

and

$$\begin{aligned}&\big\langle A_-^{g_1^{-1}}(\mathcal{Y})^p(w_1,z)w_3',w_2\big\rangle\\&=\big\langle g_1w_3',\mathcal{Y}^{-p}(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{-\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p}(z^{-1})L_{W_1}(0)}w_1,z^{-1})w_2\big\rangle\end{aligned}\tag{2.28}$$

when $\arg z=0$ and

$$\begin{aligned}&\big\langle A_+^{g_1}(\mathcal{Y})^p(w_1,z)w_3',w_2\big\rangle\\&=\big\langle w_3',\mathcal{Y}^{-p-1}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p-1}(z^{-1})L_{W_1}(0)}w_1,z^{-1}\big)g_1^{-1}w_2\big\rangle\end{aligned}\tag{2.29}$$

and

$$\begin{aligned}&\big\langle A_-^{g_1^{-1}}(\mathcal{Y})^p(w_1,z)w_3',w_2\big\rangle\\&=\big\langle g_1w_3',\mathcal{Y}^{-p-1}\big(\mathrm{e}^{zL_{W_1}(1)}\mathrm{e}^{-\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{2l_{-p-1}(z^{-1})L_{W_1}(0)}w_1,z^{-1}\big)w_2\big\rangle\end{aligned}\tag{2.30}$$

when $\arg z\neq 0$.

Let $\big(W,Y_W^g\big)$ be a $g$-twisted $V$-module. When $W_1=V$, $W_2=W_3=W$ and $\mathcal{Y}=Y_W^g$, by definition, $A_+\big(Y_W^g\big)=A_-\big(Y_W^g\big)=A_+^{1_V}\big(Y_W^g\big)=A_-^{1_V}\big(Y_W^g\big)=\big(Y_W^g\big)'$ (see Section 4).

**Theorem 2.3.** *The linear maps $A_+(\mathcal{Y})$, $A_-(\mathcal{Y})$, $A_+^{g_1}(\mathcal{Y})$ and $A_-^{g_1^{-1}}(\mathcal{Y})$ are twisted intertwining operators of types*

$$\binom{\phi_{g_1}(W_2')}{W_1W_3'},\quad\binom{W_2'}{W_1\phi_{g_1^{-1}}(W_3')},\quad\binom{\varphi_{g_1}(W_2')}{W_1W_3'}\quad\text{and}\quad\binom{W_2'}{W_1\varphi_{g_1^{-1}}(W_3')},$$

*respectively.*

*Proof.* As in the proof of Theorem 2.1, we need only prove the results for $A_+(\mathcal{Y})$ and $A_-(\mathcal{Y})$. The proofs for $A_+^{g_1}(\mathcal{Y})$ and $A_-^{g_1^{-1}}(\mathcal{Y})$ are reduced to the proofs for $A_+(\mathcal{Y})$ and $A_-(\mathcal{Y})$ using the definitions of $\varphi_{g_1}$ and $\varphi_{g_1^{-1}}$ and (2.27)–(2.30).

The proof for $A_+(\mathcal{Y})$ and $A_-(\mathcal{Y})$ is essentially the same as the proof of [17, Theorem 6.1]. But the proof here is much more complicated because the correlation functions involved are not of the explicit form as in [17]. As in [17], we prove only the

duality property since other properties are essentially the same as in the untwisted case.

We first give the multivalued analytic functions with preferred branches in the duality property. We shall denote these multivalued analytic functions for $A_+(\mathcal{Y})$ and $A_-(\mathcal{Y})$ by $h_+(z_1, z_2; u, w_1, w_3', w_2)$ and $h_-(z_1, z_2; u, w_1, w_3', w_2)$, respectively. Let $f(z_1, z_2; u, w_1, w_3', w_2)$ be the multivalued analytic function with the preferred branch $f^e(z_1, z_2; u, w_1, w_2, w_3')$ in the duality property for the twisted intertwining operator $\mathcal{Y}$. Define

$$h_\pm(z_1, z_2; u, w_1, w_3', w_2)$$

(2.31)
$$= f\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\big(-z_1^2\big)^{-L_V(0)}u, \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pm \pi \mathrm{i} L_{W_1}(0)}\big(z_2^2\big)^{-L_{W_1}(0)}w_1, w_2, w_3'\big)$$

and choose the preferred branch $h_\pm^e(z_1, z_2; u, w_1, w_3', w_2)$ of $h_\pm(z_1, z_2; u, w_1, w_3', w_2)$ as follows: On the subregion $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_1, \arg z_2 \neq 0$ of $M_0^2$, let

$$\begin{aligned} h_+^e(z_1, z_2; u, w_1, w_3', w_2) = f^{b_{13}^{-1} b_{23}^{-1}}\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z^{-1} L_V(1)}\big(-z_1^2\big)^{-L_V(0)}u, \\ \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_2) L_{W_1}(0)}w_1, w_2, w_3'\big). \end{aligned} \tag{2.32}$$

and

$$\begin{aligned} h_-^e(z_1, z_2; u, w_1, w_3', w_2) = f^{b_{13}^{-1} b_{23}^{-1} b_{12}^{-1}}\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z^{-1} L_V(1)}\big(-z_1^2\big)^{-L_V(0)}u, \\ \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{-\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_2) L_{W_1}(0)}w_1, w_2, w_3'\big). \end{aligned}$$

For general $(z_1, z_2) \in M_0^2$, we define $h_\pm^e(z_1, z_2; u, w_1, w_3', w_2)$ to be the unique analytic extensions on $M_0^2$.

Let $u \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w_3' \in W_3'$. We consider $z_1, z_2 \in \mathbb{C}$ satisfying $\big|z_2^{-1}\big| > \big|z_1^{-1}\big| > 0$ (or equivalently $|z_1| > |z_2| > 0$) and $\arg z_1, \arg z_2 \neq 0$. Since $\big|z_2^{-1}\big| > \big|z_1^{-1}\big| > 0$ and $\arg z_1, \arg z_2 \neq 0$, from (2.26), $\big(Y_{W_2}^{g_2}\big)' = A_+\big(Y_{W_2}^{g_2}\big)$ and the duality property for $\mathcal{Y}$, we know that

$$\begin{aligned} \big\langle w_3', \mathcal{Y}\big(\mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_2) L_{W_1}(0)}w_1, z_2^{-1}\big) \\ \times Y_{W_2}^{g_2}\big(\mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}g_1^{-1}u, z_1^{-1}\big)w_2\big\rangle \end{aligned} \tag{2.33}$$

is absolutely convergent and if in addition, $-\frac{3\pi}{2} < \arg\big(z_1^{-1} - z_2^{-1}\big) - \arg z_2^{-1} < -\frac{\pi}{2}$, its sum is equal to

$$\begin{aligned} &f^e\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}g_1^{-1}u, \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_2) L_{W_1}(0)}w_1, w_2, w_3'\big) \\ &\quad = f^e\big(z_1^{-1}, z_2^{-1}; g_1^{-1}\mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u, \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_2) L_{W_1}(0)}w_1, w_2, w_3'\big). \end{aligned}$$

We know that

$$\begin{aligned} &\langle \phi_{g_1}\big(\big(Y_{W_2}^{g_2}\big)'\big)(u, z_1)A_+(\mathcal{Y})(w_1, z_2)w_3', w_2\rangle \\ &\quad = \big\langle \big(Y_{W_2}^{g_2}\big)'\big(g_1^{-1}u, z_1\big)A_+(\mathcal{Y})(w_1, z_2)w_3', w_2\big\rangle \\ &= \big\langle w_3', \mathcal{Y}^{-1}\big(\mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_2) L_{W_1}(0)}w_1, z_2^{-1}\big) \\ &\quad \times \big(Y_{W_2}^{g_2}\big)^{-1}\big(\mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}g_1^{-1}u, z_1^{-1}\big)w_2\big\rangle \end{aligned} \tag{2.34}$$

can be obtained using the multivalued analytic function (2.31) on the region $\big|z_2^{-1}\big| > \big|z_1^{-1}\big| > 0$ starting from the value given by (2.33) by letting $z_1^{-1}$ go around 0 clockwise once (corresponding to $b_{13}^{-1}$), then letting $z_2^{-1}$ go around 0 clockwise once (corresponding to $b_{23}^{-1}b_{12}^{-1}$). Then (2.34) also converges absolutely on the region

$\left|z_2^{-1}\right| > \left|z_1^{-1}\right| > 0$ and if in addition, $-\frac{3\pi}{2} < \arg\left(z_1^{-1} - z_2^{-1}\right) - \arg z_2^{-1} < -\frac{\pi}{2}$, its sum is equal to

$$\begin{aligned}
&f^{b_{13}^{-1}b_{23}^{-1}b_{12}^{-1}}\left(z_1^{-1}, z_2^{-1}; g_1^{-1}\mathrm{e}^{z_1 L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}u, \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\right)\\
&\qquad = f^{b_{12}^{-1}b_{13}^{-1}b_{23}^{-1}}\left(z_1^{-1}, z_2^{-1}; g_1^{-1}\mathrm{e}^{z_1 L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}u,\right.\\
&\qquad\qquad \left.\mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\right). \qquad (2.35)
\end{aligned}$$

Using the equivariance property for $W_1$ and the convergence of (1.5) to $f^e(z_1, z_2; u, w_1, w_2, w_3')$ on the region $|z_2| > |z_1 - z_2| > 0$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$, we have

$$f^e\left(z_1, z_2; g_1^{-1}u, w_1, w_2, w_3'\right) = f^{b_{12}}(z_1, z_2; u, w_1, w_2, w_3').$$

Applying $b \in \mathrm{PB}_3$ to both sides of this equality, we obtain

$$f^b\left(z_1, z_2; g_1^{-1}u, w_1, w_2, w_3'\right) = f^{b_{12}b}(z_1, z_2; u, w_1, w_2, w_3') \qquad (2.36)$$

for $b \in \mathrm{PB}_3$. By (2.36) with $b = b_{12}^{-1}b_{13}^{-1}b_{23}^{-1}$, we see that (2.35) is equal to

$$f^{b_{13}^{-1}b_{23}^{-1}}\left(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}u, \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\right),$$

which by definition, is equal to $h_+^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$ and $\arg z_1, \arg z_2 \neq 0$.

What we need to prove is that when $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$, the sum of the left-hand side of (2.34) is equal to $h_+^e(z_1, z_2; u, w_2, w_1, w_3')$. We choose $z_1, z_2 \in -\mathbb{R}_+$ such that $|z_1| = -z_1 > -z_2 = |z_2| > 0$. Then $\left|z_2^{-1}\right| = -z_2^{-1} > -z_1^{-1} = \left|z_1^{-1}\right| > 0$. In this case, $\arg(z_1 - z_2) = \pi$, $\arg\left(z_1^{-1} - z_2^{-1}\right) = 0$, and $\arg z_1 = \arg z_2 = \arg z_1^{-1} = \arg z_2^{-1} = \pi$. Hence $|\arg(z_1 - z_2) - \arg z_1| = 0 < \frac{\pi}{2}$ and $-\frac{3\pi}{2} < 0 - \pi = \arg\left(z_1^{-1} - z_2^{-1}\right) - \arg z_2^{-1} < -\frac{\pi}{2}$. From the discussion above, for such $z_1$ and $z_2$, the sum of the left-hand side of (2.34) is equal to $h_+^e(z_1, z_2; u, w_2, w_1, w_3')$. On the other hand, such $z_1$ and $z_2$ satisfy $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$. Since both $h_+^e(z_1, z_2; u, w_2, w_1, w_3')$ and the sum of the left-hand side of (2.34) are analytic extensions of their restrictions on the subset given by $(z_1, z_2)$ for $z_1, z_2 \in -\mathbb{R}_+$ such that $|z_1| = -z_1 > -z_2 = |z_2| > 0$, the sum of the left-hand side of (2.34) is equal to $h_+^e(z_1, z_2; u, w_2, w_1, w_3')$ when $|z_1| > |z_2| > 0$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$.

Generalizing the convergence and analytic extension of (2.34) above, we can prove that

$$\left\langle \phi_{g_1}\left(\left(Y_{W_2}^{g_2}\right)'\right)(u_1, z_1)\cdots\phi_{g_1}\left(\left(Y_{W_2}^{g_2}\right)'\right)(u_{k-1}, z_{k-1})A_+(\mathcal{Y})(w_1, z_k)w_3', w_2\right\rangle \qquad (2.37)$$

is absolutely convergent on the region $|z_1| > \cdots > |z_k| > 0$ and its sum can be maximally extended to a multivalued analytic function on the region $M^k$ for $k \in \mathbb{Z}_+ + 3$, $u_1, \ldots, u_{k-1} \in V$, $w_1 \in W_1$, $w_2 \in W_2$ and $w_3' \in W_3'$. In fact, the same calculations as in (2.34) shows that (2.37) is equal to

$$\begin{aligned}
&\left\langle w_3', \mathcal{Y}^{-1}\left(\mathrm{e}^{z_k L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_k)L_{W_1}(0)}w_1, z_k^{-1}\right)\right.\\
&\quad\times\left(Y_{W_2}^{g_2}\right)^{-1}\left(\mathrm{e}^{z_{k-1}L_V(1)}\left(-z_{k-1}^{-2}\right)^{L_V(0)}g_1^{-1}u_{k-1}, z_{k-1}^{-1}\right)\cdots\\
&\quad\left.\times\left(Y_{W_2}^{g_2}\right)^{-1}\left(\mathrm{e}^{z_1 L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}g_1^{-1}u_1, z_1^{-1}\right)w_2\right\rangle \qquad (2.38)
\end{aligned}$$

when $\arg z_1, \ldots, \arg z_k \neq 0$. From the duality properties in Definitions 1.1 and 1.8, we know that

$$\begin{aligned}
&\left\langle w_3', \left(Y_{W_2}^{g_2}\right)^{-1}\left(\mathrm{e}^{z_{k-1}L_V(1)}\left(-z_{k-1}^{-2}\right)^{L_V(0)}g_1^{-1}u_{k-1}, z_{k-1}^{-1}\right)\cdots\right.\\
&\quad\times\left(Y_{W_2}^{g_2}\right)^{-1}\left(\mathrm{e}^{z_1 L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)}g_1^{-1}u_1, z_1^{-1}\right)\\
&\quad\left.\times\mathcal{Y}^{-1}\left(\mathrm{e}^{z_k L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_k)L_{W_1}(0)}w_1, z_k^{-1}\right)w_2\right\rangle \qquad (2.39)
\end{aligned}$$

is absolutely convergent on the region $\left|z_{k-1}^{-1}\right| > \cdots > \left|z_1^{-1}\right| > \left|z_k^{-1}\right| > 0$. Then for $M_{ij} \in \mathbb{Z}_+$ $(1 \le j < i \le k-1)$ such that

$$x^{M_{ij}} Y_V\big(\mathrm{e}^{z_i L_V(1)}\big(-z_i^{-2}\big)^{L_V(0)} g_1^{-1} u_i, x\big)\mathrm{e}^{z_j L_V(1)}\big(-z_j^{-2}\big)^{L_V(0)} g_1^{-1} u_j \in V[[x]],$$

we have

$$\begin{aligned}\prod_{1\le j<i\le k-1}\big(z_i^{-1}-z_j^{-1}\big)^{M_{ij}}\big\langle w_3', \big(Y_{W_2}^{g_2}\big)^{-1}\big(\mathrm{e}^{z_{k-1}L_V(1)}\big(-z_{k-1}^{-2}\big)^{L_V(0)} g_1^{-1}u_{k-1}, z_{k-1}^{-1}\big)\cdots\\ \times\big(Y_{W_2}^{g_2}\big)^{-1}\big(\mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)} g_1^{-1}u_1, z_1^{-1}\big)\\ \times\mathcal{Y}^{-1}\big(\mathrm{e}^{z_k L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_k)L_{W_1}(0)} w_1, z_k^{-1}\big)w_2\big\rangle \end{aligned} \tag{2.40}$$

is absolutely convergent on the region $\left|z_i^{-1}\right| > \left|z_k^{-1}\right| > 0$ for $i = 1, \dots, k-1$, for $i \ne j$ because $z_i^{-1} = z_j^{-1}$ for $i \ne j$ are poles of (2.39) and are canceled by $\prod_{1\le j<i\le k-1}(z_i^{-1}-z_j^{-1})^{M_{ij}}$ in (2.40). Moreover, the sum of (2.40) can be analytically extended to a maximal multivalued analytic function on $\big\{(z_1, \dots, z_k) \in \mathbb{C}^k \mid z_i \ne z_k,\ i = 1, \dots, k-1\big\}$. Using the duality property in Definition 1.8 repeatedly, we see that

$$\begin{aligned}\prod_{1\le i<j\le k-1}\big(z_i^{-1}-z_j^{-1}\big)^{M_{ij}}\big\langle w_3', \mathcal{Y}^{-1}\big(\mathrm{e}^{z_k L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_k)L_{W_1}(0)} w_1, z_k^{-1}\big)\\ \times\big(Y_{W_2}^{g_2}\big)^{-1}\big(\mathrm{e}^{z_{k-1}L_V(1)}\big(-z_{k-1}^{-2}\big)^{L_V(0)} g_1^{-1}u_{k-1}, z_{k-1}^{-1}\big)\cdots\\ \times\big(Y_{W_2}^{g_2}\big)^{-1}\big(\mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)} g_1^{-1}u_1, z_1^{-1}\big)w_2\big\rangle\end{aligned}$$

is absolutely convergent on the region $\left|z_i^{-1}\right| < \left|z_k^{-1}\right|$ for $i = 1, \dots, k-1$, $z_i^{-1} \ne z_j^{-1}$ for $i \ne j$, and its sum can be analytically extended to a maximal multivalued analytic function on $\big\{(z_1, \dots, z_k) \in \mathbb{C}^k \mid z_i \ne z_k,\ i = 1, \dots, k-1\big\}$. Thus (2.38) is absolutely convergent on the region $|z_1| > \cdots > |z_k| > 0$ and consequently (2.37) is absolutely convergent on the region $|z_1| > \cdots > |z_k| > 0$, $\arg z_1, \dots, \arg z_k \ne 0$. But the absolute convergence of (2.37) on the region $|z_1| > \cdots > |z_k| > 0$, $\arg z_1, \dots, \arg z_k \ne 0$ implies the absolute convergence of (2.37) on the region $|z_1| > \cdots > |z_k| > 0$. Moreover, its sum has a maximal analytic extension on the region $M^k$ and from the discussion above, we have

$$h(z_1, \dots, z_k) = \sum_{j_1, \dots, j_{k-1} = J'}^{J}\sum_{i=1}^{I}\sum_{m=0}^{M} z_1^{j_1}\cdots z_{k-1}^{j_{k-1}} z_k^{r_i}(\log z_k)^m f_{j_1,\dots,j_{k-1};i,m}\big(z_1^{-1}, \dots, , z_k^{-1}\big),$$

where $h(z_1, \dots, z_k)$ is the maximal analytic extension of the sum of (2.37) on the region $M^k$, $J', J \in \mathbb{Z}$, $r_1, \dots, r_I \in \mathbb{C}$, $M \in \mathbb{N}$ such that

$$\mathrm{e}^{z_l L_V(1)}\big(-z_l^{-2}\big)^{L_V(0)} g_1^{-1} u_l = \sum_{j_l = J'}^{J} z_l^{j_l} u_l^{j_l}, \qquad l = 1, \dots, k-1,$$

$$\mathrm{e}^{z_k L_{W_1}(1)}\mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)}\mathrm{e}^{-2(\log z_k)L_{W_1}(0)} w_1 = \sum_{i=1}^{I}\sum_{m=0}^{M} z_k^{r_i}(\log z_k)^m w_1^{i,m}$$

for $u_l^{j_l} \in V$ and $w_1^{i,m} \in W_1$, and $f_{j_1,\dots,j_{k-1};i,m}(z_1, \dots, z_k)$ for $J', J \in \mathbb{Z}$, $r_1, \dots, r_I \in \mathbb{C}$, $M \in \mathbb{N}$ are the maximal analytic extensions of the sums of $\big\langle w_3', Y_{W_2}^{g_2}\big(u_1^{j_1}, z_1\big)\cdots Y_{W_2}^{g_2}\big(u_{k-1}^{j_{k-1}}, z_{k-1}\big)\mathcal{Y}\big(w_1^{i,m}, z_k\big)w_2\big\rangle$. We will show at the end of this proof that a component-isolated singularity of the form $(z_1, \dots, z_{k-1}) - \beta = \delta$ for $\beta \in \mathbb{C}^{k-1}$ and $\delta \in \{0, \infty\}^k$ of any value at $\zeta = z_k$ of $\prod_{1\le i<<j\le k-1}(z_i - z_j)^{M_{ij}} h(z_1, \dots, z_{k-1}, \zeta)$ is regular.

Next we consider the product of $A_+(\mathcal{Y})$ and the twisted vertex operator $\big(Y_{W_3}^{g_2}\big)'$. Let $u$, $w_1$, $w_2$ and $w_3'$ be the same as above. When $\big|z_1^{-1}\big| > \big|z_2^{-1}\big| > 0$ (or equivalently $|z_2| > |z_1| > 0$),

$$\begin{aligned}&\big\langle w_3', Y_{W_2}^{g_2}\big(\mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u, z_1^{-1}\big)\\ &\times\mathcal{Y}\big(\mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, z_2^{-1}\big)w_2\big\rangle\end{aligned}\tag{2.41}$$

converges absolutely and if in addition, $\big|\arg\big(z_1^{-1}-z_2^{-1}\big)-\arg z_1^{-1}\big| < \frac{\pi}{2}$, its sum is equal to

$$f^e\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u, \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\big).$$

We know that when $\arg z_1, \arg z_2 \neq 0$,

$$\begin{aligned}&\big\langle A_+(\mathcal{Y})(w_1,z_2)\big(Y_{W_2}^{g_2}\big)'(u,z_1)w_3', w_2\big\rangle\\ &= \big\langle w_3', (Y_{W_2}^{g_2})^{-1}\big(\mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u, z_1^{-1}\big)\\ &\times\mathcal{Y}^{-1}\big(\mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, z_2^{-1}\big)w_2\big\rangle\end{aligned}\tag{2.42}$$

can be obtained using the multivalued analytic function (2.31) on the region $\big|z_1^{-1}\big| > \big|z_2^{-1}\big| > 0$ starting from the value given by (2.41) by letting $z_2^{-1}$ go around 0 clockwise once (corresponding to $b_{23}^{-1}$), then letting $z_1^{-1}$ go around 0 clockwise once (corresponding to $b_{12}^{-1}b_{13}^{-1}$). Then (2.42) also converges absolutely and if in addition, $\big|\arg\big(z_1^{-1}-z_2^{-1}\big)-\arg z_1^{-1}\big| < \frac{\pi}{2}$, its sum is equal to

$$\begin{aligned}&f^{b_{23}^{-1}b_{12}^{-1}b_{13}^{-1}}\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u, \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\big)\\ &= f^{b_{13}^{-1}b_{23}^{-1}b_{12}^{-1}}\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u,\\ &\mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\big).\end{aligned}\tag{2.43}$$

We now show that the sum of (2.42) is equal to (2.43). We choose $z_1, z_2 \in -\mathbb{R}_+$ such that $|z_2| = -z_2 > -z_1 = |z_1| > 0$. Then $\big|z_1^{-1}\big| = -z_1^{-1} > -z_2^{-1} = \big|z_2^{-1}\big| > 0$. In this case, $\arg(z_1-z_2) = 0$, $\arg\big(z_1^{-1}-z_2^{-1}\big) = \pi$, and $\arg z_1 = \arg z_2 = \arg z_1^{-1} = \arg z_2^{-1} = \pi$. Hence $-\frac{3\pi}{2} < 0-\pi = \arg(z_1-z_2)-\arg z_2 < -\frac{\pi}{2}$ and $|\arg\big(z_1^{-1}-z_2^{-1}\big)-\arg z_1^{-1}| = 0 < \frac{\pi}{2}$. From the discussion above, for such $z_1$ and $z_2$, the sum of (2.42) is equal to (2.43). Using analytic extension, we see that when $|z_2| > |z_1| > 0$ and $-\frac{3\pi}{2} < \arg(z_1-z_2)-\arg z_2 < -\frac{\pi}{2}$, the sum of (2.42) is equal to (2.43).

On the region $|z_2| > |z_1| > 0$, (2.43) is in fact equal to $h_+^e(z_1, z_2; u, w_2, w_1, w_3')$. This can be proved as follows: On the intersection of the region $|z_2| > |z_1| > 0$ and $M_0^2$, the branch $h_+^e(z_1, z_2; u, w_2, w_1, w_3')$ is obtained by analytically extending (2.32) defined on the intersection of the region $|z_1| > |z_2| > 0$, $|\arg(z_1-z_2)-\arg z_1| < \frac{\pi}{2}$, $\arg z_1, \arg z_2 \neq 0$ and $M_0^2$. We need to find what is this analytic extension. Let $\xi, \zeta \in -\mathbb{R}_+$ satisfying $\xi < \zeta < 0$. Then we have $|\xi| > |\zeta| > 0$, $\arg(\xi-\zeta) = \arg\xi = \arg\zeta = \pi$ so that $|\arg(\xi-\zeta)-\arg\xi| = 0 < \frac{\pi}{2}$. By definition,

$$\begin{aligned}h_+^e(\xi,\zeta;u,w_2,w_1,w_3') &= f^{b_{13}^{-1}b_{23}^{-1}}\big(\xi^{-1}, \zeta^{-1}; \mathrm{e}^{\xi^{-1}L_V(1)}\big(-\xi^2\big)^{-L_V(0)}u,\\ &\mathrm{e}^{\zeta L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log\zeta)L_{W_1}(0)}w_1, w_2, w_3'\big).\end{aligned}\tag{2.44}$$

Let $\gamma = (\gamma_1, \gamma_2)$ be the path from $(\xi,\zeta)$ to $(\zeta,\xi)$ given by the upper half circle $\gamma_1$ centered at $\frac{\xi+\zeta}{2}$ with radius $\frac{-\xi+\zeta}{2}$ from $\xi$ to $\zeta$ and the lower half circle $\gamma_2$ centered at $\frac{\xi+\zeta}{2}$ with radius $\frac{-\xi+\zeta}{2}$ from $\zeta$ to $\xi$. Since by definition, $(\xi,\zeta)$, $(\zeta,\xi)$ are in $M_0^2$ and the path $\gamma$ does not cross the boundary of $M_0^2$, $\gamma$ is a continuous path in $M_0^2$. So by definition, the value at $(z_1,z_2) = (\zeta,\xi)$ of the multivalued analytic function $h_+(z_1,z_2;u,w_2,w_1,w_3')$ obtained by starting with the value $h_+^e(\xi,\zeta;u,w_2,w_1,w_3')$ and analytically extending along $\gamma$ is $h_+^e(\zeta,\xi;u,w_2,w_1,w_3')$. On the other hand, we

need to find out what the right-hand side of (2.44) is analytically extended to along the same path $\gamma$. We have the path $\gamma' = \big(\gamma_1^{-1}, \gamma_2^{-1}\big)$ from $\big(\xi^{-1}, \zeta^{-1}\big)$ to $\big(\zeta^{-1}, \xi^{-1}\big)$, where $\gamma_1^{-1}$ and $\gamma_2^{-1}$ are the paths on the complex plane from $\xi^{-1}$ to $\zeta^{-1}$ and from $\zeta^{-1}$ to $\xi^{-1}$, respectively, given by $\gamma_1^{-1}(t) = \gamma_1(t)^{-1}$ and $\gamma_2^{-1}(t) = \gamma_1(t)^{-1}$. But $\gamma'$ is not a continuous path in $M_0^2$ because when $t$ goes from 0 to 1, $\big(\gamma_1^{-1}(t), \gamma_2^{-1}(t)\big)$ goes from $\big(\xi^{-1}, \zeta^{-1}\big)$ to $\big(\zeta^{-1}, \xi^{-1}\big)$ along the path $\gamma'$ but $\gamma_1^{-1}(t) - \gamma_2^{-1}(t)$ when $t$ goes from 0 to 1 crosses the positive real line clockwise. Crossing the positive real line clockwise corresponds to changing the branch by an action of $b_{12}^{-1}$. So we see that the value at $\big(z_1^{-1}, z_2^{-1}\big) = \big(\zeta^{-1}, \xi^{-1}\big)$ of the multivalued analytic function

(2.45)
$$f\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)} u, \mathrm{e}^{z_2 L_{W_1}(1)} \mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} w_1, w_2, w_3'\big)$$

of $z_1^{-1}$ and $z_2^{-1}$ obtained by starting with the value

$$f^{b_{13}^{-1} b_{23}^{-1}}\big(\xi^{-1}, \zeta^{-1}; \mathrm{e}^{\xi^{-1} L_V(1)}\big(-\xi^2\big)^{-L_V(0)} u, \mathrm{e}^{\zeta L_{W_1}(1)} \mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log \zeta) L_{W_1}(0)} w_1, w_2, w_3'\big)$$

and going along the path $\gamma'$ is

$$f^{b_{13}^{-1} b_{23}^{-1} b_{12}^{-1}}\big(\zeta^{-1}, \xi^{-1}; \mathrm{e}^{\zeta^{-1} L_V(1)}\big(-\zeta^2\big)^{-L_V(0)} u, \mathrm{e}^{\xi L_{W_1}(1)} \mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log \xi) L_{W_1}(0)} w_1, w_2, w_3'\big).$$

Since we have (2.44) and the path $\gamma$ is mapped to $\gamma'$ under the map $(z_1, z_2) \mapsto \big(z_1^{-1}, z_2^{-1}\big)$, the value $h_+^e(\zeta, \xi; u, w_2, w_1, w_3')$ of the multivalued analytic function $h_+(z_1, z_2; u, w_2, w_1, w_3')$ of $z_1$ and $z_2$ at the end point $(z_1, z_2) = (\zeta, \xi)$ of $\gamma$ and the value of the multivalued analytic function (2.45) of $z_1^{-1}$ and $z_2^{-1}$ at the end point $\big(z_1^{-1}, z_2^{-1}\big) = \big(\zeta^{-1}, \xi^{-1}\big)$ of $\gamma'$ must be the same, that is,

$$\begin{aligned} h_+^e(\zeta, \xi; u, w_2, w_1, w_3') = {} & f^{b_{13}^{-1} b_{23}^{-1} b_{12}^{-1}}\big(\zeta^{-1}, \xi^{-1}; \mathrm{e}^{\zeta^{-1} L_V(1)}\big(-\zeta^2\big)^{-L_V(0)} u, \\ & \mathrm{e}^{\xi L_{W_1}(1)} \mathrm{e}^{\pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log \xi) L_{W_1}(0)} w_1, w_2, w_3'\big). \end{aligned}$$

Since a single-valued branch on a subregion of $M_0^2$ of a multivalued analytic function on $M^2$ is determined uniquely by the value at one point in this subregion of $M_0^2$, we see that on the intersection of the region $|z_2| > |z_1| > 0$ and $M_0^2$,

$$\begin{aligned} h_+^e(z_1, z_2; u, w_2, w_1, w_3') = {} & f^{b_{13}^{-1} b_{23}^{-1} b_{12}^{-1}}\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1^{-1} L_V(1)}\big(-z_1^2\big)^{-L_V(0)} u, \\ & \mathrm{e}^{z_2 L_{W_1}(1)} \mathrm{e}^{\pm \pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} w_1, w_2, w_3'\big). \end{aligned} \tag{2.46}$$

Since on the region $|z_2| > |z_1| > 0$ and $\big|\arg\big(z_1^{-1} - z_2^{-1}\big) - \arg z_1^{-1}\big| < \frac{\pi}{2}$, the sum of (2.42) is equal to (2.43) (that is, the right-hand side of (2.46)), the sum of (2.42) is indeed equal to $h_+^e(z_1, z_2; u, w_2, w_1, w_3')$ on the same region.

We can prove the absolute convergence of

$$\big\langle \big(Y_{W_2}^{g_2}\big)'(u, z_1) A_-(\mathcal{Y})(w_1, z_2) w_3', w_2\big\rangle, \qquad \big\langle A_-(\mathcal{Y})(w_1, z_2) \phi_{g_1^{-1}}\big(\big(Y_{W_3}^{g_2}\big)'\big)(u, z_1) w_3', w_2\big\rangle$$

in the corresponding regions to $h_-^e(z_1, z_2; u, w_2, w_1, w_3')$ similarly by generalizing the proofs in [17] using the same method above for $A_-(\mathcal{Y})$. Here we omit the details.

Now we study the iterate of $A_\pm(\mathcal{Y})$ and the twisted vertex operator $Y_{W_1}^{g_1}$. When $\arg z_2 \neq 0$, from (2.26), we have

$$\begin{aligned} & \big\langle A_\pm(\mathcal{Y})\big(Y_{W_1}^{g_1}(u, z_1 - z_2) w_1, z_2\big) w_3', w_2\big\rangle \\ = {} & \big\langle w_3', \mathcal{Y}^{-1}\big(\mathrm{e}^{z_2 L_{W_1}(1)} \mathrm{e}^{\pm \pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} Y_{W_1}^{g_1}(u, z_1 - z_2) w_1, z_2^{-1}\big) w_2\big\rangle. \end{aligned} \tag{2.47}$$

As in [17], we have in the region $|z_2| > |z_1 - z_2| > 0$

$$\begin{aligned}
&e^{z_2 L_{W_1}(1)} \mathrm{e}^{\pm \pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} Y_{W_1}^{g_1}(u, z_1 - z_2) w_1 \\
&= Y_{W_1}^{g_1}\left(\mathrm{e}^{z_1 L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)} u, \frac{x x_0}{(x_2 + x_0) x_2}\right) \Big|_{\substack{x_0^n = \mathrm{e}^{n \log(z_1 - z_2)}, \log x_0 = \log(z_1 - z_2), x_2^n = \mathrm{e}^{n \log z_2} \\ \log x_2 = \log z_2, x^n = \mathrm{e}^{\pm n \pi \mathrm{i}}, \log x = \pm \pi \mathrm{i}}} \\
&\qquad \times \mathrm{e}^{z_2 L_{W_1}(1)} \mathrm{e}^{\pm \pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} w_1.
\end{aligned}$$

As in the proof of the first part of Theorem 2.1, we see that

$$\begin{aligned}
&\mathcal{Y}^{-1}\left(Y_{W_1}^{g_1}\left(\mathrm{e}^{z_1 L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)} u, \frac{x x_0}{(x_2 + x_0) x_2}\right) \Big|_{\substack{x_0^n = \mathrm{e}^{n \log(z_1 - z_2)}, \log x_0 = \log(z_1 - z_2), x_2^n = \mathrm{e}^{n \log z_2} \\ \log x_2 = \log z_2, x^n = \mathrm{e}^{\pm n \pi \mathrm{i}}, \log x = \pm \pi \mathrm{i}}}\right. \\
&\qquad \left. \times \mathrm{e}^{z_2 L_{W_1}(1)} \mathrm{e}^{\pm \pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} w_1, z_2^{-1}\right)
\end{aligned}$$

is an iterated series obtained by expanding one variable inside the series obtained from the iterate of $\mathcal{Y}^{-1}$ and $Y_{W_1}^{g_1}$. Using the same method as in the proof of the first part of Theorem 2.1 (including the use of Lemma 3.1), we can prove that the corresponding multiple series is absolutely convergent. Here we omit the detailed proof. In particular, we can calculate the sum of the series using any of the iterated sums associated with the multisum. Then the same calculations as those in the proof of [17, Theorem 6.1] shows that when $|z_1|, |z_2| > |z_1 - z_2| > 0$ and $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$, the right-hand side of (2.47) is equal to

$$\begin{aligned}
&\left\langle w_3', \mathcal{Y}^{-1}\left(\left(Y_{W_1}^{g_1}\right)^{m + \frac{1 \pm 1}{2}}\left(\mathrm{e}^{z_1 L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)} u, z_1^{-1} - z_2^{-1}\right)\right.\right. \\
&\qquad \left.\left. \times \mathrm{e}^{z_2 L_{W_1}(1)} \mathrm{e}^{\pm \pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} w_1, z_2^{-1}\right) w_2\right\rangle,
\end{aligned} \tag{2.48}$$

where $m \in \mathbb{Z}$ is given by

$$\log(z_1 - z_2) - \log z_1 - \log z_2 + \pi \mathrm{i} = l_{m+1}\left(z_1^{-1} - z_2^{-1}\right).$$

We know that

$$\begin{aligned}
&\left\langle w_3', \mathcal{Y}\left(Y_{W_1}^{g_1}\left(\mathrm{e}^{z_1 L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)} u, z_1^{-1} - z_2^{-1}\right) \mathrm{e}^{z_2 L_{W_1}(1)} \mathrm{e}^{\pm \pi \mathrm{i} L_{W_1}(0)}\right.\right. \\
&\qquad \left.\left. \times \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} w_1, z_2^{-1}\right) w_2\right\rangle
\end{aligned} \tag{2.49}$$

is absolutely convergent on the region $\left|z_2^{-1}\right| > \left|z_1^{-1} - z_2^{-1}\right| > 0$ and, if in addition $\left|\arg z_1^{-1} - \arg z_2^{-1}\right| < \frac{\pi}{2}$, the sum is equal to

$$f^e\left(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)} u, \mathrm{e}^{z_2 L_{W_1}(1)} \mathrm{e}^{\pm \pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} w_1, w_2, w_3'\right).$$

Since (2.48) can be obtained from (2.49) using the multivalued analytic function (2.31) on the region $\left|z_2^{-1}\right| > \left|z_1^{-1} - z_2^{-1}\right| > 0$ starting from the value given by (2.49) by letting $z_2^{-1}$ go around 0 clockwise once while $z_1^{-1} - z_2^{-1}$ is fixed (corresponding to $b_{13}^{-1} b_{23}^{-1}$ since to keep $z_1^{-1} - z_2^{-1}$ fixed, $z_1^{-1}$ must also go around 0 clockwise once) and then letting $z_1^{-1}$ go around $z_2^{-1}$ counterclockwise $m + \frac{1 \pm 1}{2}$ times (corresponding to $b_{12}^{m + \frac{1 \pm 1}{2}}$), we see that (2.48) is absolutely convergent on the region $\left|z_2^{-1}\right| > \left|z_1^{-1} - z_2^{-1}\right| > 0$ and if $\left|\arg z_1^{-1} - \arg z_2^{-1}\right| < \frac{\pi}{2}$, the sum is equal to

$$\begin{aligned}
&f^{b_{13}^{-1} b_{23}^{-1} b_{12}^{m + \frac{1 \pm 1}{2}}}\left(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\left(-z_1^{-2}\right)^{L_V(0)} u,\right. \\
&\qquad \left. \mathrm{e}^{z_2 L_{W_1}(1)} \mathrm{e}^{\pm \pi \mathrm{i} L_{W_1}(0)} \mathrm{e}^{-2(\log z_2) L_{W_1}(0)} w_1, w_2, w_3'\right).
\end{aligned}$$

In the case of $A_+$, we consider the subset $S_+$ of $(z_1, z_2) \in \mathbb{C}^2$ such that $z_1, z_2 \in -\mathbb{R}_+$, $z_1 < z_2$, and $-z_1^{-1} > z_1^{-1} - z_2^{-1} > 0$. This set is certainly nonempty since

for $z_1 = -\frac{1}{2}$, $z_2 = -\frac{1}{3}$, we have $z_1 < z_2$ and $-z_1^{-1} = 2 > 1 = z_1^{-1} - z_2^{-1} > 0$. Then for $(z_1, z_2) \in S_+$, we have

$$|z_1| > |z_2| > 0, \qquad \arg z_1 = \arg z_2 = \arg z_1^{-1} = \arg z_2^{-1} = \arg(z_1 - z_2) = \pi,$$
$$\arg\big(z_1^{-1} - z_2^{-1}\big) = 0, \qquad |\arg(z_1 - z_2) - \arg z_1| = 0 < \frac{\pi}{2},$$
$$\big|z_2^{-1}\big| = -z_2^{-1} > -z_1^{-1} > z_1^{-1} - z_2^{-1} = \big|z_1^{-1} - z_2^{-1}\big| > 0, \qquad \big|\arg z_1^{-1} - \arg z_2^{-1}\big| = 0 < \frac{\pi}{2},$$

and

$$\begin{aligned}
&\log(z_1 - z_2) - \log z_1 - \log z_2 + \pi\mathrm{i} \\
&= \log|z_1 - z_2| + \arg(z_1 - z_2)\mathrm{i} - \log|z_1| - \arg z_1\mathrm{i} - \log|z_2| - \arg z_2\mathrm{i} + \pi\mathrm{i} \\
&= \log\left|\frac{z_1 - z_2}{z_1 z_2}\right| = \log\big|z_1^{-1} - z_2^{-1}\big| + \arg\big(z_1^{-1} - z_2^{-1}\big) \\
&= \log\big(z_1^{-1} - z_2^{-1}\big).
\end{aligned}$$

In particular, for $(z_1, z_2) \in S_+$, we have $m+1 = 0$. Then on $S_+$, (2.48) is absolutely convergent to

$$\begin{aligned}
&f^{b_{13}^{-1}b_{23}^{-1}}\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u, \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\big) \\
&\qquad = h_+^e(z_1, z_2; u, w_1, w_3', w_2).
\end{aligned}$$

By analytic extension, we see that the sum of (2.48) is equal to $h_+^e(z_1, z_2; u, w_1, w_3', w_2)$ on the subregion of $M_0^2$ given by $\big|z_2^{-1}\big| > \big|z_1^{-1} - z_2^{-1}\big| > 0$, $\big|\arg z_1^{-1} - \arg z_2^{-1}\big| < \frac{\pi}{2}$. By analytic extension again, we see that the sum of (2.47) is equal to $h_+^e(z_1, z_2; u, w_1, w_3', w_2)$ on the subregion of $M_0^2$ given by $|z_2| > |z_1 - z_2| > 0$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$.

In the case of $A_-$, we consider the subset $S_-$ of $(z_1, z_2) \in \mathbb{C}^2$ such that $z_1, z_2 \in -\mathbb{R}_+$, $z_1 < z_2$, and $-z_1^{-1} > z_1^{-1} - z_2^{-1} > 0$. This set is nonempty since for $z_1 = -\frac{1}{2}$, $z_2 = -\frac{1}{3}$, we have $z_1 < z_2$ and $-z_1^{-1} = 2 > 1 = -2 + 3 = z_1^{-1} - z_2^{-1} > 0$. Then for for $(z_1, z_2) \in S_-$, we have $|z_1| > |z_2| > 0$, $|\arg(z_1 - z_2) - \arg z_1| = 0 < \frac{\pi}{2}$, $\arg(z_1 - z_2) = \pi \geq \pi$, $\big|z_2^{-1}\big| = -z_2^{-1} > -z_1^{-1} > z_1^{-1} - z_2^{-1} = \big|z_1^{-1} - z_2^{-1}\big| > 0$, $\big|\arg z_1^{-1} - \arg z_2^{-1}\big| = 0 < \frac{\pi}{2}$, and

$$\log(z_1 - z_2) - \log z_1 - \log z_2 + \pi\mathrm{i} = \log\left|\frac{z_1 - z_2}{z_1 z_2}\right|\mathrm{i} = \log\big(z_1^{-1} - z_2^{-1}\big).$$

In particular, for $(z_1, z_2) \in S_-$, we also have $m + 1 = 0$. Then on $S_-$, (2.48) is absolutely convergent to

$$\begin{aligned}
&f^{b_{13}^{-1}b_{23}^{-1}b_{12}^{-1}}\big(z_1^{-1}, z_2^{-1}; \mathrm{e}^{z_1 L_V(1)}\big(-z_1^{-2}\big)^{L_V(0)}u, \mathrm{e}^{z_2 L_{W_1}(1)}\mathrm{e}^{-\pi\mathrm{i}L_{W_1}(0)}\mathrm{e}^{-2(\log z_2)L_{W_1}(0)}w_1, w_2, w_3'\big) \\
&\qquad = h_-^e(z_1, z_2; u, w_1, w_3', w_2).
\end{aligned}$$

By analytic extension, we see that the sum of (2.48) is equal to $h_-^e(z_1, z_2; u, w_1, w_3', w_2)$ on the subregion of $M_0^2$ given by $\big|z_2^{-1}\big| > \big|z_1^{-1} - z_2^{-1}\big| > 0$, $\big|\arg z_1^{-1} - \arg z_2^{-1}\big| < \frac{\pi}{2}$. By analytic extension again, we see that the sum of (2.47) is equal to $h_-^e(z_1, z_2; u, w_1, w_3', w_2)$ on the subregion of $M_0^2$ given by $|z_2| > |z_1 - z_2| > 0$, $|\arg z_1 - \arg z_2| < \frac{\pi}{2}$.

Finally we prove that for any fixed $z_k \in \mathbb{C}^\times$, a component-isolated singularity of the form $(z_1, \dots, z_{k-1}) - \beta = \delta$ for $\beta \in \mathbb{C}^{k-1}$ and $\delta \in \{0, \infty\}^k$ of any value at $\zeta = z_k$ of $\prod_{1 \leq i < \leq j \leq k-1}(z_i - z_j)^{M_{ij}} h(z_1, \dots, z_{k-1}, \zeta)$ is regular. Generalizing the proof above, we can show that

$$\big\langle \phi_{g_1}\big(\big(Y_{W_2}^{g_2}\big)'\big)(u_1, z_1) \cdots \phi_{g_1}\big(\big(Y_{W_2}^{g_2}\big)'\big)(u_{k-2}, z_{k-2}) A_\pm(\mathcal{Y})(Y_{W_1}^{g_1}(u, z_{k-1} - z_k)w_1, z_k)w_3', w_2\big\rangle$$

is absolutely convergent on the region $|z_1| > \cdots > |z_{k-2}| > |z_k| > |z_{k-1} - z_k| > 0$ and the maximal analytic extension of its sum on $M^k$ is $h(z_1, \dots, z_k)$. But we have proved above that (2.37) is absolutely convergent on the region $|z_1| > \cdots >$

$|z_k| > 0$ and the maximum analytic extension of its sum is also $h(z_1, \dots, z_k)$. Thus $\prod_{1\le i<j\le k-1}(z_i - z_j)^{M_{ij}} h(z_1, \dots, \zeta)$ can be expanded as

$$\prod_{1\le i<j\le k-1}(z_i - z_j)^{M_{ij}} \langle \phi_{g_1}\big(\big(Y_{W_2}^{g_2}\big)'\big)(u_1, z_1)\cdots\phi_{g_1}\big(\big(Y_{W_2}^{g_2}\big)'\big)(u_{k-1}, z_{k-1})$$

$$\times A_\pm(\mathcal{Y})\big(Y_{W_1}^{g_1}(u, z_{k-1} - z_k)w_1, z_k\big)w_3', w_2\rangle, \tag{2.50}$$

a series in powers of $z_1, \dots, z_{k-2}, z_{k-1} - z_k$ and finitely many integral powers of $\log z_1, \dots, \log z_{k-2}, \log(z_{k-1} - z_k)$. From the expansion (2.50), we see that the component-isolated singularity $(z_1, \dots, z_{k-2}, z_{k-1} - z_k) = (\infty, \dots, \infty, 0)$ of the value $\prod_{1\le i<j\le k-1}(z_i - z_j)^{M_{ij}} h(z_1, \dots, z_k)$ at $\zeta = z_k$ is regular. Using the commutativity for $A_\pm(\mathcal{Y})$ (which is already proved), we see that the component-isolated singularity $(z_1, \dots, z_{k-2}, z_{k-1} - z_k) = (0, \dots, 0)$ of the value $\prod_{1\le i<j\le k-1}(z_i - z_j)^{M_{ij}} h(z_1, \dots, z_k)$ at $\zeta = z_k$ is also regular. The regularity of the other component-isolated singularity of the form above can be proved either similarly or easily by using the relation between $h(z_1, \dots, z_k)$ and $f_{j_1,\dots,j_{k-1};i,m}(z_1, \dots, z_k)$. □

## 3. A convergence lemma

Let $A$ be a finite subset of $\mathbb{C}/\mathbb{Z}$, $R_\mu \in \mu$ for $\mu \in A$, $D$ a subset of $\cup_{\mu\in A}(R_\mu - \mathbb{N})$, $\Delta \in \mathbb{C}$ and $a_{n,j,i} \in \mathbb{C}$ for $n \in D$, $j = 0, \dots, M$ and $i = 1, \dots, N$. Consider the triple series

$$\sum_{n\in D}\sum_{j=0}^{M}\sum_{i=0}^{N} a_{n,j,i} \mathrm{e}^{(-\Delta+n+1)\log z_1}(\log z_1)^j \mathrm{e}^{(-n-1)\log z_2}(\log z_2)^i \tag{3.1}$$

for $z_1, z_2 \in \mathbb{C}^\times$.

For any $z_1$, $z_2$ satisfying $|z_1| > |z_2|$, $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$, we have

$$\mathrm{e}^{\alpha\log(z_1-z_2)} = \sum_{k\in\mathbb{N}} \binom{\alpha}{k} (-1)^k \mathrm{e}^{(\alpha-k)\log z_1} z_2^k, \tag{3.2}$$

$$\log(z_1 - z_2) = \log z_1 + \sum_{k\in\mathbb{Z}_+} \frac{(-1)^k}{k} z_1^{-k} z_2^k, \tag{3.3}$$

and for $z_2 \ne 0$, we have $\mathrm{e}^{\alpha l_{q_2}(-z_2)} = \mathrm{e}^{\alpha\pi\mathrm{i}}\mathrm{e}^{\alpha\log z_2}$, $l_{q_2}(-z_2) = \log z_2 + \pi\mathrm{i}$, where $\alpha \in \mathbb{C}$ and $q_2 = 0, 1$ if $\arg z_2 < \pi$, $\arg z_2 \ge \pi$, respectively. Note that in our notations, $\log z = l_0(z)$ for $z \in \mathbb{C}^\times$.

For $n \in D$, $j = 0, \dots, M$, $k \in \mathbb{Z}_{\ge 0}$, $s = 0, \dots, j$, define $b_{n,j,k,s} \in \mathbb{C}$ as the coefficients of the following formal power series expansion

$$(x + y)^{-\Delta+n+1}\log(x + y)^j = \sum_{k\in\mathbb{N}}\sum_{s=0}^{j} b_{n,j,k,s} x^{-\Delta+n+1-k} y^k (\log x)^s, \tag{3.4}$$

where $x$ and $y$ are formal variables. From (3.2), (3.3) and (3.4), when $|z_1| > |z_2|$ and $|\arg(z_1 - z_2) - \arg z_1| < \frac{\pi}{2}$, we have the following expansion:

$$\mathrm{e}^{(-\Delta+n+1)\log(z_1-z_2)}(\log(z_1 - z_2))^j$$

$$= \sum_{k\in\mathbb{N}}\sum_{s=0}^{j} (-1)^k b_{n,j,k,s} \mathrm{e}^{(-\Delta+n+1-k)\log z_1} z_2^k (\log z_1)^s. \tag{3.5}$$

Now consider

(3.6)

$$\sum_{n\in D}\sum_{j=0}^{M}\sum_{i=0}^{N} a_{n,j,i} \mathrm{e}^{(-\Delta+n+1)\log(z_1-z_2)}(\log(z_1 - z_2))^j \mathrm{e}^{(-n-1)\log(-z_2)}(\log(-z_2))^i.$$

Using (3.5), we can further expand each term in the right-hand side of (3.6) so that the right-hand side of (3.6) becomes the iterated sum

$$\begin{aligned}&\sum_{n\in D}\sum_{j=0}^{M}\sum_{i=0}^{N}a_{n,j,i}\left(\sum_{k\in\mathbb{N}}\sum_{s=0}^{j}(-1)^k b_{n,j,k,s}\mathrm{e}^{(-\Delta+n+1-k)\log z_1}z_2^k(\log z_1)^s\right)\\&\qquad\times\mathrm{e}^{(-n-1)\log(-z_2)}(\log(-z_2))^i\\&=\sum_{n\in D}\sum_{j=0}^{M}\sum_{i=0}^{N}\left(\sum_{k\in\mathbb{N}}\sum_{s=0}^{j}a_{n,j,i}(-1)^k b_{n,j,k,s}\mathrm{e}^{(-\Delta+n+1-k)\log z_1}z_2^k(\log z_1)^s\right.\\&\qquad\left.\times\ \mathrm{e}^{(-n-1)\log(-z_2)}(\log(-z_2))^i\right),\end{aligned}\tag{3.7}$$

where the inner sum is absolutely convergent on the region $|z_1|>|z_2|>0$.

We are interested in the convergence of the multisum

$$\begin{aligned}&\sum_{n\in D}\sum_{j=0}^{M}\sum_{i=0}^{N}\sum_{k\in\mathbb{N}}\sum_{s=0}^{j}a_{n,j,i}(-1)^k b_{n,j,k,s}\mathrm{e}^{(-\Delta+n+1-k)\log z_1}z_2^k\\&\qquad\times(\log z_1)^s\mathrm{e}^{(-n-1)\log(-z_2)}(\log(-z_2))^i\end{aligned}\tag{3.8}$$

and the corresponding series

$$\begin{aligned}&\sum_{m\in D-\mathbb{N}}\sum_{s=0}^{M}\sum_{i=0}^{N}\left(\sum_{j=s}^{M}\sum_{\substack{n-k=m\\ n\in D,k\in\mathbb{N}}}a_{n,j,i}b_{n,j,k,s}\right)\mathrm{e}^{(-\Delta+m+1)\log z_1}\\&\qquad\times(\log z_1)^s\mathrm{e}^{(-m-1)\log(-z_2)}(\log(-z_2))^i\end{aligned}\tag{3.9}$$

in powers of $z_1$ and $z_2$ and nonnegative integral powers of $\log z_1$ and $\log z_2$.

**Lemma 3.1.** *Assume that the triple series* (3.1) *and the series obtained from* (3.1) *by taking derivatives of each term in* (3.1) *with respect to $z_1$ and $z_2$ are absolutely convergent on the region given by $|z_1|>|z_2|>0$. Then the multisum* (3.8) *is absolutely convergent on the region $|z_1|>2|z_2|>0$. Assume in addition that* (3.1) *is convergent on the region $|z_1|>|z_2|>0$, $|\arg(z_1-z_2)-\arg z_1|<\frac{\pi}{2}$ to a single-valued analytic branch $f^e(z_1,z_2)$ on $M_0^2$ of a maximally extended multivalued analytic function on $M^2$ such that $f^e(z_1-z_2,-z_2)$ has no singular point in the region $|z_1|>|z_2|>0$. Then the iterated sum* (3.9) *is absolutely convergent on the region $|z_1|>|z_2|>0$, $|\arg z_1-\arg(z_1-z_2)|<\frac{\pi}{2}$ to $f^e(z_1-z_2,-z_2)$.*

*Proof.* Let $\tilde{n}=R_\mu-n$. Then the sum $\sum_{n\in D}$ in (3.1) can be written as the same as $\sum_{\mu\in A}\sum_{\tilde{n}\in\mathbb{N}}$. So the series (3.1) can be written as

$$\sum_{\mu\in A}\sum_{\tilde{n}\in\mathbb{N}}\sum_{j=0}^{M}\sum_{i=0}^{N}a_{R_\mu-\tilde{n},j,i}\mathrm{e}^{(-\Delta+R_\mu-\tilde{n}+1)\log z_1}(\log z_1)^j\mathrm{e}^{(-R_\mu+\tilde{n}-1)\log z_2}(\log z_2)^i.$$

By assumption, the series

$$\sum_{\mu\in A}\sum_{\tilde{n}\in\mathbb{N}}\sum_{j=0}^{M}\sum_{i=0}^{N}a_{R_\mu-\tilde{n},j,i}\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\mathrm{e}^{(-\Delta+R_\mu-\tilde{n}+1)\log z_1}(\log z_1)^j\mathrm{e}^{(-R_\mu+\tilde{n}-1)\log z_2}(\log z_2)^i.\tag{3.10}$$

for $p, q \in \mathbb{N}$ is absolutely convergent on the region $|z_1| > |z_2| > 0$. For any $r > 1$, consider

$$\sum_{\mu\in A}\sum_{\tilde{n}\in\mathbb{N}}\sum_{j=0}^{M}\sum_{i=0}^{N} a_{R_\mu-\tilde{n},j,i}\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\mathrm{e}^{(-\Delta+R_\mu-\tilde{n}+1)\log z_1}(\log z_1)^j \times\mathrm{e}^{(-R_\mu+\tilde{n}-1)\log z_2}(\log z_2)^i r^{\tilde{n}}. \tag{3.11}$$

Then the absolute convergence of (3.10) on the region $|z_1| > |z_2| > 0$ is equivalent to the absolute convergence of (3.11) on the region $|z_1| > r|z_2| > 0$. But the absolute convergence of (3.11) on the region $|z_1| > r|z_2| > 0$ is in turn equivalent to the absolute convergence of

$$\sum_{\tilde{n}\in\mathbb{N}} a_{R_\mu-\tilde{n},j,i}\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\left(\frac{z_2}{z_1}\right)^{\tilde{n}} r^{\tilde{n}} \tag{3.12}$$

on the same region $|z_1| > r|z_2| > 0$. But as a power series in $\frac{z_2}{z_1}$, (3.12) with $p, q = 0$ has a removable singularity at $\frac{z_2}{z_1} = 0$. Then we see that (3.12) with $p, q = 0$ is absolutely convergent on the region $|\frac{z_2}{z_1}| < \frac{1}{r}$ and is hence also uniformly convergent on the closed region $|\frac{z_2}{z_1}| \le r_2$ for any positive $r_2 < \frac{1}{r}$. In particular, by the property of power series, (3.12) for $p, q \in \mathbb{N}$ is also absolutely convergent on the region $|\frac{z_2}{z_1}| < \frac{1}{r}$ and is uniformly convergent on the closed region $|\frac{z_2}{z_1}| \le r_2$. Thus for such $r_2$, (3.11) is a (finite) linear combination of the uniformly convergent series (3.12) on the region $|\frac{z_2}{z_1}| \le r_2$ with analytic functions of $z_1$ and $z_2$ on $M_0^2$ as coefficients.

Substituting $z_1 - z_2$ and $-z_2$ for $z_1$ and $z_2$, respectively, in (3.11), we see that

$$\sum_{\mu\in A}\sum_{\tilde{n}\in\mathbb{N}}\sum_{j=0}^{M}\sum_{i=0}^{N} a_{n,j,i}(-1)^q\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\mathrm{e}^{(-\Delta+n+1)\log(z_1-z_2)}(\log(z_1-z_2))^j \times\mathrm{e}^{(-n-1)\log(-z_2)}(\log(-z_2))^i r^{\tilde{n}} \tag{3.13}$$

is absolutely convergent on the region $|z_1 - z_2| > r|z_2| > 0$ and is a (finite) linear combination of uniformly convergent series on the region $|\frac{z_2}{z_1-z_2}| \le r_2$ with analytic coefficients on $M_0^2$ for any positive $r_2 < \frac{1}{r}$. Now we expand each term in (3.13) using (3.5) and then passing the derivative $\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}$ to each term since the right-hand side of (3.5) is essentially a power series in $\frac{z_2}{z_1}$ to obtain the iterated sum

$$\begin{aligned}
&\sum_{\mu\in A}\sum_{\tilde{n}\in\mathbb{N}}\sum_{j=0}^{M}\sum_{i=0}^{N} a_{R_\mu-\tilde{n},j,i}\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\left(\left(\sum_{k\in\mathbb{N}}\sum_{s=0}^{j}(-1)^k b_{R_\mu-\tilde{n},j,k,s}\right.\right.\\
&\left.\left.\times\mathrm{e}^{(-\Delta+R_\mu-\tilde{n}+1-k)\log z_1} z_2^k(\log z_1)^s\right)\mathrm{e}^{(-R_\mu+\tilde{n}-1)\log(-z_2)}(\log(-z_2))^i r^{\tilde{n}}\right)\\
=&\sum_{\mu\in A}\sum_{\tilde{n}\in\mathbb{N}}\sum_{j=0}^{M}\sum_{i=0}^{N}\left(\sum_{k\in\mathbb{N}}\sum_{s=0}^{j} a_{R_\mu-\tilde{n},j,i}(-1)^k b_{R_\mu-\tilde{n},j,k,s}\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\mathrm{e}^{(-\Delta+R_\mu-\tilde{n}+1-k)\log z_1}\right.\\
&\left.\times\ z_2^k(\log z_1)^s\mathrm{e}^{(-R_\mu+\tilde{n}-1)\log(-z_2)}(\log(-z_2))^i r^{\tilde{n}}\right),
\end{aligned} \tag{3.14}$$

where the inner sum is absolutely convergent on the region $|z_1| > |z_2| > 0$. Then as a subseries of (3.14) divided by a suitable function of the form $\mathrm{e}^{\lambda\log z_1}\mathrm{e}^{\mu\log(-z_2)}(\log(-z_2))^\nu$

for some $\lambda, \mu \in \mathbb{C}$, $\nu \in \mathbb{N}$, the series

$$\sum_{\tilde{n}\in\mathbb{N}}\left(\sum_{k\in\mathbb{N}}\sum_{s=0}^{j}(-1)^{\tilde{n}+k}a_{R_\mu-\tilde{n},j,i}b_{R_\mu-\tilde{n},j,k,s}\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\left(\frac{z_2}{z_1}\right)^{\tilde{n}+k}(\log z_1)^s\right)r^{\tilde{n}}$$

$$=\sum_{\tilde{n}\in\mathbb{N}}\left(\sum_{\tilde{k}\in\mathbb{N}}\sum_{s=0}^{j}(-1)^{\tilde{k}}a_{R_\mu-\tilde{n},j,i}b_{R_\mu-\tilde{n},j,\tilde{k}-\tilde{n},s}\frac{\partial^{p+q}}{\partial z_1^p\partial z_2^q}\left(\frac{z_2}{z_1}\right)^{\tilde{k}}(\log z_1)^s\right)r^{\tilde{n}}, \tag{3.15}$$

where $b_{R_\mu-\tilde{n},j,\tilde{k}-\tilde{n},s}$ is defined to be 0 when $\tilde{k}<\tilde{n}$, is also absolutely convergent on the region $|z_1-z_2|>r|z_2|$ and is uniformly convergent on the closed region $\left|\frac{z_2}{z_1-z_2}\right|\le r_2$ for any positive $r_2<\frac{1}{r}$, where the inner sum is absolutely convergent in the region $|z_1|>|z_2|$. In the region $|z_1|>(1+r)|z_2|>0$, we have $|z_1-z_2|\ge |z_1|-|z_2|>r|z_2|>0$. Then (3.15) is absolutely convergent on the region $|z_1|>(1+r)|z_2|$ or $|\frac{z_2}{z_1}|<\frac{1}{1+r}$ with the inner sum absolutely convergent in the region $|z_1|>|z_2|$ or $|\frac{z_2}{z_1}|<1$. Note that for fixed $\frac{z_2}{z_1}$, $z_1$ can be any complex number and thus $\log z_1$ can also be any nonzero complex number whose imaginary part is in $[0,2\pi)$. Moreover, (3.15) is convergent absolutely on the same region when $\log z_1$ is replaced by any choice of branches of the logarithm of $z_1$. This means that $\zeta_1=\frac{z_2}{z_1}$ and $\zeta_2=\log z_1$ can be viewed as independent variables and, after the change of variables from $z_1$, $z_2$ to $\zeta_1$ and $\zeta_2$,

$$\sum_{\tilde{n}\in\mathbb{N}}\left(\sum_{\tilde{k}\in\mathbb{N}}\sum_{s=0}^{j}(-1)^{\tilde{k}}a_{R_\mu-\tilde{n},j,i}b_{R_\mu-\tilde{n},j,\tilde{k}-\tilde{n},s}\left(-\zeta_1\mathrm{e}^{\zeta_2}\frac{\partial}{\partial\zeta_1}+\mathrm{e}^{-\zeta_2}\frac{\partial}{\partial\zeta_2}\right)^p\right.$$

$$\left.\times\left(\mathrm{e}^{-\zeta_2}\frac{\partial}{\partial\zeta_1}\right)^q\zeta_1^{\tilde{k}}\zeta_2^s\right)\zeta_3^{\tilde{n}} \tag{3.16}$$

is absolutely convergent on the region given by $|\zeta_1|<\frac{1}{1+r}$, $\zeta_2\in\mathbb{C}$ and $|\zeta_3|<r$. From (3.16), we see that

$$\sum_{\tilde{n}\in\mathbb{N}}\left(\sum_{\tilde{k}\in\mathbb{N}}\sum_{s=0}^{j}(-1)^{\tilde{k}}a_{R_\mu-\tilde{n},j,i}b_{R_\mu-\tilde{n},j,\tilde{k}-\tilde{n},s}\frac{\partial^{\tilde{p}+\tilde{q}}}{\partial\zeta_1^{\tilde{p}}\partial\zeta_2^{\tilde{q}}}\zeta_1^{\tilde{k}}\zeta_2^s\right)\zeta_3^{\tilde{n}} \tag{3.17}$$

for $\tilde{p},\tilde{q}\in\mathbb{N}$ is absolutely convergent on the region given by $|\zeta_1|<\frac{1}{1+r}$, $\zeta_2\in\mathbb{C}$ and $|\zeta_3|<r$.

On the other hand, on the region $|\frac{z_2}{z_1}|\le\frac{r_2}{1+r_2}$ for positive $r_2<\frac{1}{r}$,

$$\left|\frac{z_2}{z_1-z_2}\right|\le\frac{|z_2|}{|z_1|-|z_2|}\le\frac{|z_2|}{\frac{1+r_2}{r_2}|z_2|-|z_2|}=r_2.$$

Then for $\frac{z_2}{z_1}\in\mathbb{C}$ satisfying $|\frac{z_2}{z_1}|\le\frac{r_2}{1+r_2}$ for positive $r_2<\frac{1}{r}$ and $\log z_1\in\mathbb{C}$, (3.15) is uniformly convergent with the inner sum also uniformly convergent on the same closed region. Thus (3.17) is uniformly convergent on the region $|\zeta_1|\le\frac{r_2}{1+r_2}$, $\zeta_2\in\mathbb{C}$ and $|\zeta_3|\le r$ with the inner sum also uniformly convergent on the same closed region. We have shown that the series (3.17) for $\tilde{p},\tilde{q}\in\mathbb{N}$ obtained by taking derivatives of each term in the series (3.17) with $\tilde{p},\tilde{q}=0$ with respect to $\zeta_1$, $\zeta_2$ and $\zeta_3$ is uniformly convergent on the region $|\zeta_1|\le\frac{r_2}{1+r_2}$, $\zeta_2\in\mathbb{C}$ and $|\zeta_3|\le r$. In particular, the derivatives of (3.17) with $\tilde{p},\tilde{q}=0$ with respect to $\zeta_1$, $\zeta_2$ and $\zeta_3$ exist and is equal to the sum of the series obtained by taking the corresponding derivatives of each term in (3.17) with $\tilde{p},\tilde{q}=0$ on the region $|\zeta_1|<\frac{r_2}{1+r_2}$, $\zeta_2\in\mathbb{C}$ and $|\zeta_3|<r$. Then the sum of (3.17) with $\tilde{p},\tilde{q}=0$ is an analytic function of $\zeta_1$, $\zeta_2$ and $\zeta_3$ on the same open region.

For $r > 1$, let $\zeta_1$, $\zeta_2$, $\zeta_3$ be complex numbers satisfying $|\zeta_1| < \frac{1}{(1+r)r}$, $\zeta_2 \in \mathbb{C}$ and $|\zeta_3| < r$. Then we have $\frac{|\zeta_1|}{1-|\zeta_1|} < (1+r)|\zeta_1| < \frac{1}{r}$. We choose $r_2$ such that $\frac{|\zeta_1|}{1-|\zeta_1|} < r_2 < (1+r)|\zeta_1|$. From $\frac{|\zeta_1|}{1-|\zeta_1|} < r_2$, we obtain $|\zeta_1| < \frac{r_2}{1+r_2}$. We also have $0 < r_2 < (1+r)|\zeta_1| < \frac{1}{r}$. Now $\zeta_1$, $\zeta_3$ satisfy $|\zeta_1| \le \frac{r_2}{1+r_2}$ and $|\zeta_3| < r$. This means that the sum of (3.17) with $\tilde{p}, \tilde{q} = 0$ is analytic and the derivatives can be calculated term by term at $\zeta_1$, $\zeta_2$, $\zeta_3$. So the sum of (3.17) with $\tilde{p}, \tilde{q} = 0$ is analytic and the derivatives can be calculated term by term on the polydisc $|\zeta_1| < \frac{1}{(1+r)r}$, $\zeta_2 \in \mathbb{C}$ and $|\zeta_3| < r$ for any $r > 1$. Since analytic functions on polydiscs can be expanded as power series, the sum of (3.17) with $\tilde{p}, \tilde{q} = 0$ can be expanded as a power series in $\zeta_1$, $\zeta_2$ and $\zeta_3$ and the coefficients of the power series expansion can be obtained using its derivatives with respect to $\zeta_1$, $\zeta_2$ and $\zeta_3$ evaluated at $\zeta_1 = \zeta_2 = \zeta_3 = 0$. By taking the derivatives term by term, we see that the coefficients of the power series expansion of this analytic function are equal to the coefficients $(-1)^{\tilde{k}} a_{R_\mu - \tilde{n}, j, i} b_{R_\mu - \tilde{n}, j, \tilde{k} - \tilde{n}, s}$ of the iterated series (3.17) with $\tilde{p}, \tilde{q} = 0$. Since the power series expansion of an analytic function is an absolutely convergent multisum, we see that the triple series

$$\sum_{\tilde{n} \in \mathbb{N}} \sum_{\tilde{k} \in \mathbb{N}} \sum_{s=0}^{j} (-1)^{\tilde{k}} a_{R_\mu - \tilde{n}, j, i} b_{R_\mu - \tilde{n}, j, \tilde{k} - \tilde{n}, s} \zeta_1^{\tilde{k}} \zeta_2^{s} \zeta_3^{\tilde{n}} \tag{3.18}$$

is absolutely convergent on the region $|\zeta_1| < \frac{1}{(1+r)r}$, $\zeta_2 \in \mathbb{C}$ and $|\zeta_3| < r$.

In particular, on the region $|z_1| > (1+r)r|z_2|$, taking $\zeta_1 = \frac{z_2}{z_1}$, $\zeta_2 = \log z_1$ and $\zeta_3 = 1$ in (3.18), we see that the triple series

$$\sum_{\tilde{n} \in \mathbb{N}} \sum_{\tilde{k} \in \mathbb{N}} \sum_{s=0}^{j} (-1)^{\tilde{k}} a_{R_\mu - \tilde{n}, j, i} b_{R_\mu - \tilde{n}, j, \tilde{k} - \tilde{n}, s} \left( \frac{z_2}{z_1} \right)^{\tilde{k}} (\log z_1)^s \tag{3.19}$$

is absolutely convergent. Since $r$ is an arbitrary real number satisfying $r > 1$, (3.19) is in fact absolutely convergent on the region $|z_1| > 2|z_2|$. Multiplying $\mathrm{e}^{(-\Delta + R_\mu + 1)\log z_1} \mathrm{e}^{(-R_\mu - 1)\log(-z_2)} \times (\log(-z_2))^i$ to (3.19) and summing over $A$, $j = 0, \dots, M$ and $i = 0, \dots, N$, we see that the multiseries

$$\begin{aligned}
&\sum_{\mu \in A} \sum_{\tilde{n} \in \mathbb{N}} \sum_{j=0}^{M} \sum_{i=0}^{N} \sum_{\tilde{k} \in \mathbb{N}} \sum_{s=0}^{j} (-1)^{\tilde{k}} a_{R_\mu - \tilde{n}, j, i} b_{R_\mu - \tilde{n}, j, \tilde{k} - \tilde{n}, s} \mathrm{e}^{(-\Delta + R_\mu + 1)\log z_1} \\
&\qquad \times \mathrm{e}^{(-R_\mu - 1)\log(-z_2)} (\log(-z_2))^i \left( \frac{z_2}{z_1} \right)^{\tilde{k}} (\log z_1)^s \\
&= \sum_{n \in D} \sum_{j=0}^{M} \sum_{i=0}^{N} \sum_{k \in \mathbb{N}} \sum_{s=0}^{j} a_{n,j,i} (-1)^k b_{n,j,k,s} \mathrm{e}^{(-\Delta + n + 1 - k)\log z_1} z_2^k (\log z_1)^s \\
&\qquad \times \mathrm{e}^{(-n-1)\log(-z_2)} (\log(-z_2))^i
\end{aligned} \tag{3.20}$$

is absolutely convergent on the region $|z_1| > 2|z_2| > 0$, proving the first part of the lemma.

In the case that the additional assumption holds, from the proof above and the additional assumption, the multisum (3.20), which is equal to the iterated series in the right-hand side of (3.7), is absolutely convergent on the region $|z_1| > 2|z_2| > 0$, $|\arg z_1 - \arg(z_1 - z_2)| < \frac{\pi}{2}$ to $f^e(z_1 - z_2, -z_2)$. In particular, (3.9) as an iterated sum of (3.20) is also absolutely convergent on the region $|z_1| > 2|z_2| > 0$, $|\arg z_1 - \arg(z_1 - z_2)| < \frac{\pi}{2}$ to $f^e(z_1 - z_2, -z_2)$. Since there is no singular point of $f^e(z_1 - z_2, -z_2)$ in the region $|z_1| > |z_2| > 0$, (3.9) must also be absolutely convergent when $|z_1| > |z_2| > 0$ and is thus absolutely convergent on the region $|z_1| > |z_2| > 0$, $|\arg z_1 - \arg(z_1 - z_2)| < \frac{\pi}{2}$ to $f^e(z_1 - z_2, -z_2)$. □

**Remark 3.2.** There is a subtlety about the convergence regions for (3.20) and (3.9). Note that in general the multisum (3.20) might not be absolutely convergent on the larger region $|z_1| > |z_2| > 0$. This is because (3.20) is not a series in powers of $z_1$ and $z_2$ and nonnegative integral powers of $\log z_1$ and $\log z_2$. Even if we restore the variable $\zeta_3$ in the proof of the lemma above to obtain a series in powers of $z_1$, $z_2$, $\zeta_3$ and nonnegative integral powers of $\log z_1$ and $\log z_2$, since we do not have the assumption that the sum of this series can be analytically extended to a region containing $|z_1| > |z_2| > 0$, this series might not be absolutely convergent on any region containing $|z_1| > |z_2| > 0$.

**Remark 3.3.** Lemma 3.1 can be generalized to the case of more than two variables. Consider the series

$$\sum_{\substack{n_1,\dots,n_k\in D\\ n_1+\cdots+n_k=\Delta-k}} \sum_{j_1,\dots,j_{k-1}=0}^{M} \sum_{i=0}^{N} a_{n_1,\dots,n_k,j_1,\dots,j_{k-1},i} \mathrm{e}^{(-n_1-1)\log z_1} (\log z_1)^{j_1} \cdots$$
$$\times \mathrm{e}^{(-n_{k-1}-1)\log z_l} (\log z_l)^{j_{k-1}} \mathrm{e}^{(-n_k-1)\log z_k} (\log z_k)^i \tag{3.21}$$

for $z_1, \dots, z_k \in \mathbb{C}^\times$. Replacing $z_i$ for $i = 1, \dots, k-1$ and $z_k$ by $z_i - z_k$ and $-z_k$, respectively, we obtain the following series:

$$\sum_{\substack{n_1,\dots,n_k\in D\\ n_1+\cdots+n_k=\Delta-k}} \sum_{j_1,\dots,j_{k-1}=0}^{M} \sum_{i=0}^{N} a_{n_1,\dots,n_k,j_1,\dots,j_{k-1},i} \mathrm{e}^{(-n_1-1)\log(z_1-z_k)} \log(z_1-z_k)^{j_1} \cdots$$
$$\times \mathrm{e}^{(-n_{k-1}-1)\log(z_{k-1}-z_k)} (\log(z_{k-1}-z_k))^{j_{k-1}} \mathrm{e}^{(-n_k-1)\log(-z_k)} (\log(-z_k))^i. \tag{3.22}$$

Expanding $\mathrm{e}^{(-n_1-1)\log(z_1-z_k)} \log(z_1-z_k)^{j_1}, \dots, \mathrm{e}^{(-n_{k-1}-1)\log(z_{k-1}-z_k)} (\log(z_{k-1}-z_k))^{j_l}$ in (3.22) using (3.5) with $-\Delta+n+1$ replaced by $-n_1-1, \dots, -n_l-1$, respectively, we obtain an iterated series. This iterated series also has the corresponding multisum and other iterated series. A generalization of Lemma 3.1 says that if (3.21) is absolutely convergent on the region $|z_1| > \cdots > |z_k| > 0$ and its sum is equal to a suitable single-valued branch of a multivalued analytic function on a suitable subregion, then the iterated series obtained by taking the sums in (3.22) first and then taking the sums given by the expansions of $\mathrm{e}^{(-n_1-1)\log(z_1-z_k)}(\log(z_1-z_k)^{j_1}, \dots, \mathrm{e}^{(-n_{k-1}-1)\log(z_{k-1}-z_k)}(\log(z_{k-1}-z_k))^{j_l}$ is absolutely convergent on the region $|z_1| > \cdots > |z_k| > 0$ and its sum is equal to a given single-valued branch on a suitable subregion. Here we omit the details of this generalization and its proof.

## References


[1] Carnahan S., Miyamoto M., Regularity of fixed-point vertex operator subalgebras, arXiv:1603.05645.

[2] Dixon L., Harvey J.A., Vafa C., Witten E., Strings on orbifolds, *Nuclear Phys. B* **261** (1985), 678–686.

[3] Dixon L., Harvey J.A., Vafa C., Witten E., Strings on orbifolds. II, *Nuclear Phys. B* **274** (1986), 285–314.

[4] Du J., A cyclic orbifold theory for holomorphic vertex operator algebras and applications, Ph.D. Thesis, Rutgers University, 2026.

[5] Frenkel I.B., Huang Y.-Z., Lepowsky J., On axiomatic approaches to vertex operator algebras and modules, *Mem. Amer. Math. Soc.* **104** (1993), viii+64 pages.

[6] Frenkel I.B., Lepowsky J., Meurman A., A natural representation of the Fischer–Griess Monster with the modular function $J$ as character, *Proc. Nat. Acad. Sci. USA* **81** (1984), 3256–3260.

[7] Frenkel I.B., Lepowsky J., Meurman A., Vertex operator calculus, in Mathematical Aspects of String Theory (San Diego, Calif., 1986), *Adv. Ser. Math. Phys.*, Vol. 1, World Scientific Publishing, Singapore, 1987, 150–188.

[8] Frenkel I.B., Lepowsky J., Meurman A., Vertex operator algebras and the Monster, *Pure Appl. Math.*, Vol. 134, Academic Press Boston, MA, 1988.
[9] Huang Y.-Z., A theory of tensor products for module categories for a vertex operator algebra. IV, *J. Pure Appl. Algebra* **100** (1995), 173–216, arXiv:q-alg/9505019.
[10] Huang Y.-Z., Generalized rationality and a "Jacobi identity" for intertwining operator algebras, *Selecta Math. (N.S.)* **6** (2000), 225–267, arXiv:q-alg/9704008.
[11] Huang Y.-Z., Differential equations and intertwining operators, *Commun. Contemp. Math.* **7** (2005), 375–400, arXiv:math/0206206.
[12] Huang Y.-Z., Differential equations, duality and modular invariance, *Commun. Contemp. Math.* **7** (2005), 649–706, arXiv:math/0303049.
[13] Huang Y.-Z., Rigidity and modularity of vertex tensor categories, *Commun. Contemp. Math.* **10** (2008), 871–911, arXiv:math/0502533.
[14] Huang Y.-Z., Vertex operator algebras and the Verlinde conjecture, *Commun. Contemp. Math.* **10** (2008), 103–154, arXiv:math/0406291.
[15] Huang Y.-Z., Generalized twisted modules associated to general automorphisms of a vertex operator algebra, *Comm. Math. Phys.* **298** (2010), 265–292, arXiv:0905.0514.
[16] Huang Y.-Z., Some open problems in mathematical two-dimensional conformal field theory, in Lie Algebras, Vertex Operator Algebras, and Related Topics, *Contemp. Math.*, Vol. 695, American Mathematical Society, Providence, RI, 2017, 123–138, arXiv:1606.04493.
[17] Huang Y.-Z., Intertwining operators among twisted modules associated to not-necessarily-commuting automorphisms, *J. Algebra* **493** (2018), 346–380, arXiv:1702.05845.
[18] Huang Y.-Z., A construction of lower-bounded generalized twisted modules for a grading-restricted vertex (super)algebra, *Comm. Math. Phys.* **377** (2020), 909–945, arXiv:1903.00737.
[19] Huang Y.-Z., Representation theory of vertex operator algebras and orbifold conformal field theory, in Lie Groups, Number Theory, and Vertex Algebras, *Contemp. Math.*, Vol. 768, American Mathematical Society, Providence, RI, 2021, 221–252, arXiv:2004.01172.
[20] Huang Y.-Z., On the applicability of logarithmic tensor category theory, *New York J. Math.* **31** (2025), 902–910, arXiv:1702.00133.
[21] Huang Y.-Z., $C_1$-cofiniteness and vertex tensor categories, arXiv:2509.20737.
[22] Huang Y.-Z., Cofiniteness and $P(z)$-tensor product bifunctors in orbifold theories associated to abelian but not-necessarily-finite groups, arXiv:2601.12834.
[23] Huang Y.-Z., Lepowsky J., Tensor products of modules for a vertex operator algebra and vertex tensor categories, in Lie Theory and Geometry, *Progr. Math.*, Vol. 123, Birkhäuser, Boston, MA, 1994, 349–383, arXiv:hep-th/9401119.
[24] Huang Y.-Z., Lepowsky J., A theory of tensor products for module categories for a vertex operator algebra. I, *Selecta Math. (N.S.)* **1** (1995), 699–756, arXiv:hep-th/9309076.
[25] Huang Y.-Z., Lepowsky J., A theory of tensor products for module categories for a vertex operator algebra. II, *Selecta Math. (N.S.)* **1** (1995), 757–786, arXiv:hep-th/9309159.
[26] Huang Y.-Z., Lepowsky J., A theory of tensor products for module categories for a vertex operator algebra. III, *J. Pure Appl. Algebra* **100** (1995), 141–171, arXiv:q-alg/9505018.
[27] Huang Y.-Z., Lepowsky J., Zhang L., Logarithmic tensor category theory for generalized modules for a conformal vertex algebra, III: Intertwining maps and tensor product bifunctors, arXiv:1012.4197.
[28] Huang Y.-Z., Lepowsky J., Zhang L., Logarithmic tensor category theory for generalized modules for a conformal vertex algebra, IV: Constructions of tensor product bifunctors and the compatibility conditions, arXiv:1012.4198.
[29] Huang Y.-Z., Lepowsky J., Zhang L., Logarithmic tensor category theory for generalized modules for a conformal vertex algebra, V: Convergence condition for intertwining maps and the corresponding compatibility condition, arXiv:1012.4199.
[30] Huang Y.-Z., Lepowsky J., Zhang L., Logarithmic tensor category theory for generalized modules for a conformal vertex algebra, VI: Expansion condition, associativity of logarithmic intertwining operators, and the associativity isomorphisms, arXiv:1012.4202.
[31] Kirillov Jr. A., Modular categories and orbifold models, *Comm. Math. Phys.* **229** (2002), 309–335, arXiv:math/0104242.
[32] Kirillov Jr. A., Modular categories and orbifold models II, arXiv:math/0110221.
[33] Kirillov Jr. A., On $G$-equivariant modular categories, arXiv:math/0401119.
[34] McRae R., Twisted modules and $G$-equivariantization in logarithmic conformal field theory, *Comm. Math. Phys.* **383** (2021), 1939–2019, arXiv:1910.13226.
[35] Möller S., A cyclic orbifold theory for holomorphic vertex operator algebras and applications, Ph.D. Thesis, Technische Universität Darmstadt, 2016, arXiv:1611.09843.
[36] Moore G., Seiberg N., Classical and quantum conformal field theory, *Comm. Math. Phys.* **123** (1989), 177–254.

[37] Müger M., Galois extensions of braided tensor categories and braided crossed $G$-categories, *J. Algebra* **277** (2004), 256–281, arXiv:math/0209093.
[38] Müger M., Conformal orbifold theories and braided crossed $G$-categories, *Comm. Math. Phys.* **260** (2005), 727–762, arXiv:math/0403322.
[39] Reshetikhin N., Turaev V.G., Invariants of 3-manifolds via link polynomials and quantum groups, *Invent. Math.* **103** (1991), 547–597.
[40] Turaev V.G., Modular categories and 3-manifold invariants, *Internat. J. Modern Phys. B* **6** (1992), 1807–1824.
[41] Turaev V.G., Homotopy field theory in dimension 3 and crossed group-categories, arXiv:math/0005291.
[42] van Ekeren J., Möller S., Scheithauer N.R., Construction and classification of holomorphic vertex operator algebras, *J. Reine Angew. Math.* **759** (2020), 61–99, arXiv:1507.08142.